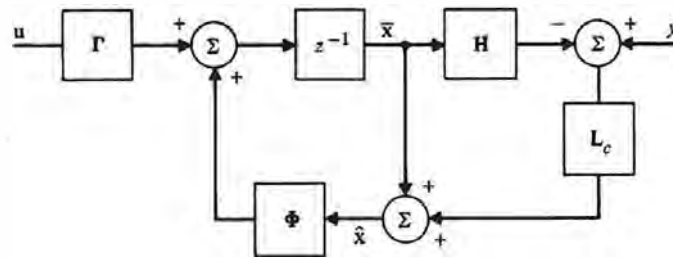


Figure 8.6
Estimator block diagram



substitutions Φ^T for Φ and $\Phi^T H^T$ for Γ yield L_c instead of K . Alternatively, we could compute L_p using Eq. (8.30) and then compute L_c using Eq. (8.37); that is

$$L_c = \Phi^{-1} L_p. \quad (8.40)$$

◆ **Example 8.6** *Current Estimator Pole Placement with MATLAB*

Design a current estimator for the satellite attitude-control system as in Examples 8.4 and 8.5. Place the z -plane poles at $z = 0.4 \pm j0.4$ and compare the error response to an initial error in the velocity with that obtained in Example 8.4.

Solution. The MATLAB statements

```
T = .1
Phi = [1 T; 0 1]
Gam = [T^2/2; T]
p = [.4+i*.4; .4-i*.4]
Lc = acker(Phi', Phi'*H', p)
```

result in

$$L_c = \begin{bmatrix} 0.68 \\ 5.2 \end{bmatrix}.$$

Therefore, the estimator implementation using Eq. (8.33) in a way that reduces the computation delay as much as possible is, before sampling

$$\begin{aligned} \bar{x}_1(k) &= \hat{x}_1(k-1) + 0.005 u(k-1) + 0.1 \hat{x}_2(k-1), \\ \bar{x}_2(k) &= \hat{x}_2(k-1) + 0.1 u(k-1), \\ x'_1 &= (1 - 0.68) \bar{x}_1(k), \\ x'_2 &= (1 - 5.2) \bar{x}_1(k) + \bar{x}_2, \end{aligned}$$

and after sampling $y(k)$

$$\begin{aligned} \hat{x}_1(k) &= x'_1 + 0.68 y(k), \\ \hat{x}_2(k) &= x'_2 + 5.2 y(k), \end{aligned}$$

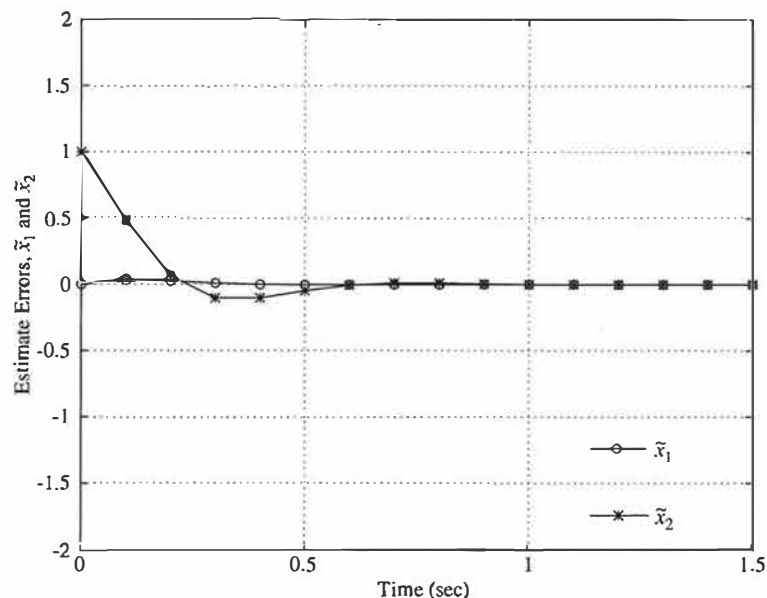
at which time the state vector estimate is available for the calculation of the control $u(k)$.

Figure 8.7 shows the time history of the estimator error equation from Eq. (8.38), again with an initial error of 0 for the position estimate and 1 rad/sec for the velocity estimate. The figure shows very similar results compared to the prediction estimator in Fig. 8.5; however, the current estimator implementation exhibits a response which is about a cycle faster.

If Φ is singular, as can happen with systems having time delays, neither Eqs. (8.39) nor (8.40) can be used. However, estimators can be designed for these systems as discussed in Section 8.6.

Note that we now have two estimates that could be used for control purposes, the predicted estimate [$\hat{\mathbf{x}}(k)$ from Eq. (8.23)] and the current estimate [$\hat{\mathbf{x}}(k)$ from Eq. (8.33)]. The current estimate is the obvious choice because it is based on the most current value of the measurement, y . Its disadvantage is that it is out of date before the computer can complete the computation of Eqs. (8.33) and (8.5), thus creating a delay that is not accounted for in the design process, which will cause less damping in the implemented system than specified by the desired poles. The use of the predicted estimate for control eliminates the modeling error from the latency because it can be calculated using the measurement, $y(k-1)$, thus providing an entire sample period to complete the necessary calculations of $u(k)$ before its value is required. Generally, however, one should use the

Figure 8.7
Time history of current
estimator error



current estimate because it provides the fastest response to unknown disturbances or measurement errors and thus better regulation of the desired output. Any deficiencies in the system response due to the latency from the computation lag that is found by simulation or experiment can be patched up with additional iterations on the desired pole locations or accounted for exactly by including computation delay in the plant model.

8.2.5 Reduced-Order Estimators

The estimators discussed so far are designed to reconstruct the entire state vector, given measurements of some of the state elements.⁸ One might therefore ask: Why bother to reconstruct the state elements that are measured directly? The answer is: You don't have to, although, when there is significant noise on the measurements, the estimator for the full state vector provides smoothing of the measured elements as well as reconstruction of the unmeasured state elements.

To pursue an estimator for only the unmeasured part of the state vector, let us partition the state vector into two parts: x_a is the portion directly measured, which is y , and x_b is the remaining portion to be estimated. The complete system description, like Eq. (8.3), becomes

$$\begin{bmatrix} x_a(k+1) \\ x_b(k+1) \end{bmatrix} = \begin{bmatrix} \Phi_{aa} & \Phi_{ab} \\ \Phi_{ba} & \Phi_{bb} \end{bmatrix} \begin{bmatrix} x_a(k) \\ x_b(k) \end{bmatrix} + \begin{bmatrix} \Gamma_a \\ \Gamma_b \end{bmatrix} u(k), \quad (8.41)$$

$$y(k) = [I \quad 0] \begin{bmatrix} x_a(k) \\ x_b(k) \end{bmatrix} \quad (8.42)$$

and the portion describing the dynamics of the unmeasured state elements is

$$x_b(k+1) = \Phi_{bb}x_b(k) + \underbrace{\Phi_{ba}x_a(k) + \Gamma_b u(k)}_{\text{known "input"}}, \quad (8.43)$$

where the right-hand two terms are known and can be considered as an input into the x_b dynamics. If we reorder the x_a portion of Eq. (8.41), we obtain

$$\underbrace{x_a(k+1) - \Phi_{aa}x_a(k) - \Gamma_a u(k)}_{\text{known "measurements"}} = \Phi_{ab}x_b(k). \quad (8.44)$$

Note that this is a relationship between a measured quantity on the left and the unknown state vector on the right. Therefore, Eqs. (8.43) and (8.44) have the same relationship to the state vector x_b that the original equation, (8.3), had to the

⁸ Reduced-order estimators (or observers) were originally proposed by Luenberger (1964). This development follows Gopinath (1971).

entire state vector $\mathbf{x}$. Following this reasoning, we arrive at the desired estimator by making the following substitutions

$$\begin{aligned}\mathbf{x} &\leftarrow \mathbf{x}_b, \\ \Phi &\leftarrow \Phi_{bb}, \\ \Gamma u(k) &\leftarrow \Phi_{ba}x_a(k) + \Gamma_b u(k), \\ y(k) &\leftarrow x_a(k+1) - \Phi_{aa}x_a(k) - \Gamma_a u(k), \\ \mathbf{H} &\leftarrow \Phi_{ab},\end{aligned}$$

into the prediction estimator equations (8.23). Thus the reduced-order estimator equations are

$$\hat{\mathbf{x}}_b(k+1) = \Phi_{bb}\hat{\mathbf{x}}_b(k) + \Phi_{ba}x_a(k) + \Gamma_b u(k) + \mathbf{L}_r[x_a(k+1) - \Phi_{aa}x_a(k) - \Gamma_a u(k) - \Phi_{ab}\hat{\mathbf{x}}_b(k)]. \quad (8.45)$$

Subtracting Eq. (8.45) from (8.43) yields the error equation

$$\tilde{\mathbf{x}}_b(k+1) = [\Phi_{bb} - \mathbf{L}_r\Phi_{ab}]\tilde{\mathbf{x}}_b(k), \quad (8.46)$$

and therefore $\mathbf{L}_r$ is selected exactly as before, that is, (a) by picking roots of

$$|z\mathbf{I} - \Phi_{bb} + \mathbf{L}_r\Phi_{ab}| = \alpha_e(z) = 0 \quad (8.47)$$

to be in desirable locations, or (b) using Ackermann's formula

$$\mathbf{L}_r = \alpha_e(\Phi_{bb}) \begin{bmatrix} \Phi_{ab} \\ \Phi_{ab}\Phi_{bb} \\ \Phi_{ab}\Phi_{bb}^2 \\ \vdots \\ \Phi_{ab}\Phi_{bb}^{n-2} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}. \quad (8.48)$$

We note here that Gopinath (1971) proved that if a full-order estimator as given by Eq. (8.23) exists, then the reduced-order estimator given by Eq. (8.45) also exists; that is, we can place the roots of Eq. (8.47) anywhere we choose by choice of $\mathbf{L}_r$.

◆ Example 8.7 Reduced-Order Estimator for Satellite Attitude

Determine a reduced-order estimator for the same case as in Examples 8.4 and 8.5.

Solution. We start out by partitioning the plant equations to fit the mold of Eqs. (8.41) and (8.42). This results in

$$\begin{aligned}\begin{bmatrix} \Phi_{aa} & \Phi_{ab} \\ \Phi_{ba} & \Phi_{bb} \end{bmatrix} &= \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} \Gamma_a \\ \Gamma_b \end{bmatrix} = \begin{bmatrix} T^2/2 \\ T \end{bmatrix} = \begin{bmatrix} 0.005 \\ 0.1 \end{bmatrix} \\ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} \text{the measured position state } y, \\ \text{the velocity to be estimated} \end{bmatrix}, \quad (8.49)\end{aligned}$$

where Φ_{aa} , and so on, are all scalars. Therefore L_r is a scalar also, and there is only one estimator pole to pick, the pole corresponding to the speed at which the estimate of scalar velocity converges. From Eq. (8.47) we pick L_r from

$$z - 1 + L_r T = 0.$$

For this estimator to be about the same speed as the two previous estimator examples, which had two poles at $z = 0.4 \pm j0.4$, we will pick the pole at $z = 0.5$; therefore $L_r T - 1 = -0.5$ and $L_r = 5$. The estimator equation, (8.45), is

$$\begin{aligned} \hat{x}_b(k) = & \hat{x}_b(k-1) + 0.1 u(k-1) \\ & + 5.0 [y(k) - y(k-1) - 0.005 u(k-1) - (0.1) \hat{x}_b(k-1)]. \end{aligned} \quad (8.50)$$

The implementation in a control computer would, before sampling, look something like

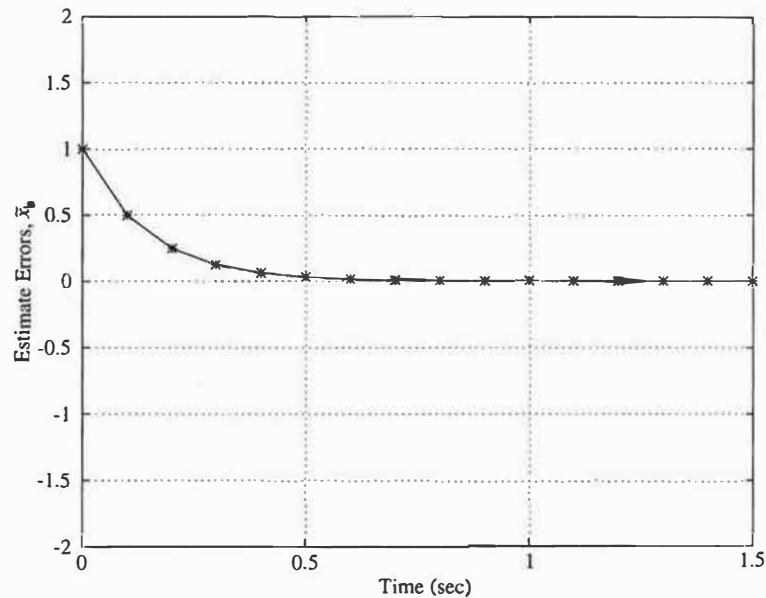
$$x' = 0.5 \hat{x}_b(k-1) + 0.075 u(k-1) - 5 y(k-1),$$

and after sampling

$$\hat{x}_b(k) = x' + 5 y(k).$$

Figure 8.8 shows the time history of the estimator-error equation (8.46) with an initial (velocity) estimate error of 1 rad/sec. The figure shows very similar results compared to the velocity element estimates in Figs. 8.5 and 8.7. Of course, there is no position estimate because this formulation assumes that the measurement is used directly without smoothing.

Figure 8.8
Time history of
reduced-order estimator
error



8.3 Regulator Design: Combined Control Law and Estimator

If we take the control law (Section 8.1) and implement it, using an estimated state vector (Section 8.2), the control system can be completed. A schematic of such a system is shown in Fig. 8.9. However, because we designed the control law assuming that the true state, $\mathbf{x}$, was fed back instead of $\hat{\mathbf{x}}$ or $\bar{\mathbf{x}}$, it is of interest to examine what effect this has on the system dynamics. We will see that it has no effect! The poles of the complete system consisting of the estimator feeding the control law will have the same poles as the two cases analyzed separately.

8.3.1 The Separation Principle

The control is now

$$u(k) = -\mathbf{K} \bar{\mathbf{x}}(k)$$

and the controlled plant equation (8.6) becomes

$$\mathbf{x}(k+1) = \Phi \mathbf{x}(k) - \Gamma \mathbf{K} \bar{\mathbf{x}}(k), \quad (8.51)$$

which can also be written in terms of the estimator error using Eq. (8.21)

$$\mathbf{x}(k+1) = \Phi \mathbf{x}(k) - \Gamma \mathbf{K}(\mathbf{x}(k) - \bar{\mathbf{x}}(k)). \quad (8.52)$$

Combining this with the estimator-error equation (8.24)⁹ we obtain two coupled equations that describe the behavior of the complete system¹⁰

$$\begin{bmatrix} \bar{\mathbf{x}}(k+1) \\ \mathbf{x}(k+1) \end{bmatrix} = \begin{bmatrix} \Phi - \mathbf{L}_p \mathbf{H} & 0 \\ \Gamma \mathbf{K} & \Phi - \Gamma \mathbf{K} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{x}}(k) \\ \mathbf{x}(k) \end{bmatrix}. \quad (8.53)$$

The characteristic equation is

$$\begin{vmatrix} z\mathbf{I} - \Phi + \mathbf{L}_p \mathbf{H} & 0 \\ \Gamma \mathbf{K} & z\mathbf{I} - \Phi + \Gamma \mathbf{K} \end{vmatrix} = 0 \quad (8.54)$$

which, because of the zero matrix in the upper right, can be written as

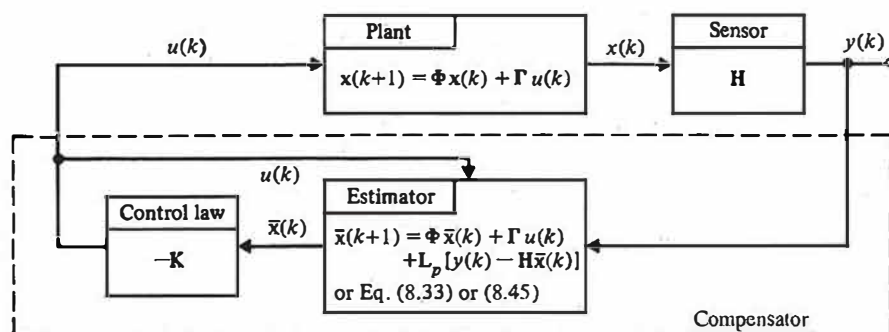
$$\left| z\mathbf{I} - \Phi + \mathbf{L}_p \mathbf{H} \right| \left| z\mathbf{I} - \Phi + \Gamma \mathbf{K} \right| = \alpha_c(z) \alpha_e(z) = 0. \quad (8.55)$$

In other words, the characteristic poles of the complete system consist of the combination of the estimator poles and the control poles that are unchanged from those obtained assuming actual state feedback. The fact that the combined control-estimator system has the same poles as those of the control alone and the

⁹ We show only the prediction estimator case. The other estimators lead to identical conclusions.

¹⁰ This description of the entire system does not apply if the estimator model is imperfect; see Section 11.5 for an analysis of that case.

Figure 8.9
Estimator and controller mechanization



estimator alone is a special case of the separation principle by which control and estimation can be designed separately yet used together.

compensator

To compare this method of design to the methods discussed in Chapter 7, we note from Fig. 8.9 that the portion within the dashed line corresponds to classical compensation. The difference equation for this system or “state-space designed compensator” is obtained by including the control feedback (because it is part of the compensation) in the estimator equations. Using Eq. (8.23) yields for the prediction estimator

$$\begin{aligned}\bar{\mathbf{x}}(k) &= [\Phi - \Gamma\mathbf{K} - \mathbf{L}_p\mathbf{H}]\bar{\mathbf{x}}(k-1) + \mathbf{L}_p y(k-1), \\ u(k) &= -\mathbf{K}\bar{\mathbf{x}}(k),\end{aligned}\quad (8.56)$$

and using Eq. (8.33) yields for the current estimator

$$\begin{aligned}\hat{\mathbf{x}}(k) &= [\Phi - \Gamma\mathbf{K} - \mathbf{L}_c\mathbf{H}\Phi + \mathbf{L}_c\mathbf{H}\Gamma\mathbf{K}]\hat{\mathbf{x}}(k-1) + \mathbf{L}_c y(k), \\ u(k) &= -\mathbf{K}\hat{\mathbf{x}}(k).\end{aligned}\quad (8.57)$$

The poles of the compensators above are obtained from, for Eq. (8.56),

$$|z\mathbf{I} - \Phi + \Gamma\mathbf{K} + \mathbf{L}_p\mathbf{H}| = 0 \quad (8.58)$$

and, for Eq. (8.57),

$$|z\mathbf{I} - \Phi + \Gamma\mathbf{K} + \mathbf{L}_c\mathbf{H}\Phi - \mathbf{L}_c\mathbf{H}\Gamma\mathbf{K}| = 0, \quad (8.59)$$

and are *neither* the control law poles, Eq. (8.8), nor the estimator poles, Eq. (8.25). These poles need not be determined during a state-space design effort, but can be of interest for comparison with compensators designed using the transform methods of Chapter 7.

If desired, Eq. (8.56) can be converted to transfer function-form using the same steps that were used in arriving at Eq. (4.64). This results in what was called

compensation in Chapter 7 and usually referred to as $D(z)$. For the prediction estimator, we find

$$\frac{U(z)}{Y(z)} = D_p(z) = -\mathbf{K}[z\mathbf{I} - \Phi + \Gamma\mathbf{K} + \mathbf{L}_p\mathbf{H}]^{-1}\mathbf{L}_p. \quad (8.60)$$

The $D(z)$ could also be found from Eq. (8.56) by using `tf.m` in MATLAB or, for the current estimator, from Eq. (8.57). Likewise, the transfer function for the reduced-order compensator is found by using the measured part of the state, x_a , directly in the control law and the estimated part, $\hat{x}_b(k)$, for the remainder. Thus, the control gain $\mathbf{K}$ needs to be partitioned, so that

$$u(k) = [K_a \quad K_b] \begin{bmatrix} x_a \\ \hat{x}_b \end{bmatrix} \quad \text{where} \quad \mathbf{K} = [K_a \quad K_b]. \quad (8.61)$$

estimator pole selection

In the previous sections, we developed techniques to compute $\mathbf{K}$ and $\mathbf{L}_p$ (which define the compensation), given the desired locations of the roots of the characteristic equations of the control and the estimator. We now know that these desired root locations will be the closed-loop system poles. The same meter sticks that applied to the classical design and were discussed in Section 7.2 also apply to picking these poles. In practice, when measurement noise is not an issue, it is convenient to pick the control poles to satisfy the performance specifications and actuator limitations, and then to pick the estimator poles somewhat faster (by a factor of 2 to 4) so that the total response is dominated by the response due to the slower control poles. It does not cost anything in terms of actuator hardware to increase the estimator gains (and hence speed of response) because they appear only in the computer. The upper limit to estimator speed of response is based on the behavior of sensor-noise rejection, which is the subject of Chapter 9.

In order to evaluate the full system response with the estimator in the loop, it is necessary to simulate both the real system and the estimator system as was formulated in Eq. (8.53). However, it is easier to see what is going on by using $\tilde{\mathbf{x}}$ or $\hat{\mathbf{x}}$ in place of $\tilde{\mathbf{x}}$. The result for the predictor case using Eq. (8.56) is

$$\begin{bmatrix} \mathbf{x}(k+1) \\ \tilde{\mathbf{x}}(k+1) \end{bmatrix} = \begin{bmatrix} \Phi & -\Gamma\mathbf{K} \\ \mathbf{L}_p\mathbf{H} & \Phi - \Gamma\mathbf{K} - \mathbf{L}_p\mathbf{H} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \tilde{\mathbf{x}}(k) \end{bmatrix}, \quad (8.62)$$

the result for the current estimator using Eq. (8.57) is

$$\begin{bmatrix} \mathbf{x}(k+1) \\ \hat{\mathbf{x}}(k+1) \end{bmatrix} = \begin{bmatrix} \Phi & -\Gamma\mathbf{K} \\ \mathbf{L}_c\mathbf{H}\Phi & \Phi - \Gamma\mathbf{K} - \mathbf{L}_c\mathbf{H}\Phi \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \hat{\mathbf{x}}(k) \end{bmatrix}, \quad (8.63)$$

and the result for the reduced-order estimator using Eq. (8.45) is

$$\begin{bmatrix} \mathbf{x}(k+1) \\ \hat{\mathbf{x}}_b(k+1) \end{bmatrix} = \begin{bmatrix} \Phi - \Gamma \begin{bmatrix} K_a & 0 \end{bmatrix} & -\Gamma\mathbf{K}_b \\ \mathbf{A} & \mathbf{B} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \hat{\mathbf{x}}_b(k) \end{bmatrix}, \quad (8.64)$$

where

$$\mathbf{A} = \mathbf{L}_r\mathbf{H}\Phi + \Phi_{ba}\mathbf{H} - \Gamma_b\mathbf{K}_a\mathbf{H} - \mathbf{L}_r\Phi_{aa}\mathbf{H}$$

and

$$\mathbf{B} = \Phi_{bb} - \Gamma_b \mathbf{K}_b - \mathbf{L}_r \Phi_{ab}$$

and where $\mathbf{K}_a$ and $\mathbf{K}_b$ are partitions of $\mathbf{K}$ according to the dimensions of x_a and x_b .

◆ Example 8.8 Compensation Based on the Predictor Estimator

Put together the full feedback control system based on the calculations already done in Examples 8.1 and 8.4, that is, using the prediction estimator. Use the control gain, $\mathbf{K} = [10 \ 3.5]$, and the estimator gain, $\mathbf{L}_p^T = [1.2 \ 5.2]$. Determine the $D(z)$ for comparison with a classical lead compensation. Plot the response of the system variables for an initial plant velocity of -1 rad/sec and zero for all other initial conditions. Comment on whether the responses are consistent with your expectations.

Solution. The compensation equations consist of Eq. (8.56) with the values of $\mathbf{K}$ and $\mathbf{L}_p$ plugged in and, being in difference-equation form, can be coded directly in a control computer. To find the transfer function form, we use `zp.m` (or Eq. (8.60)), to find that¹¹

$$D_p(z) = -30.4 \frac{z - 0.825}{z - 0.2 \pm j0.557}. \quad (8.65)$$

There is a compensator zero near the two plant poles at $z = +1$ and there are two compensator poles considerably to the left. This is very similar to a classical lead compensator except that it has two poles instead of one. State-space design using a full-order estimator will always produce compensation that is the same order as the plant. Note that the difference equation that results from this $D(z)$ will have a one cycle delay between the input and output.

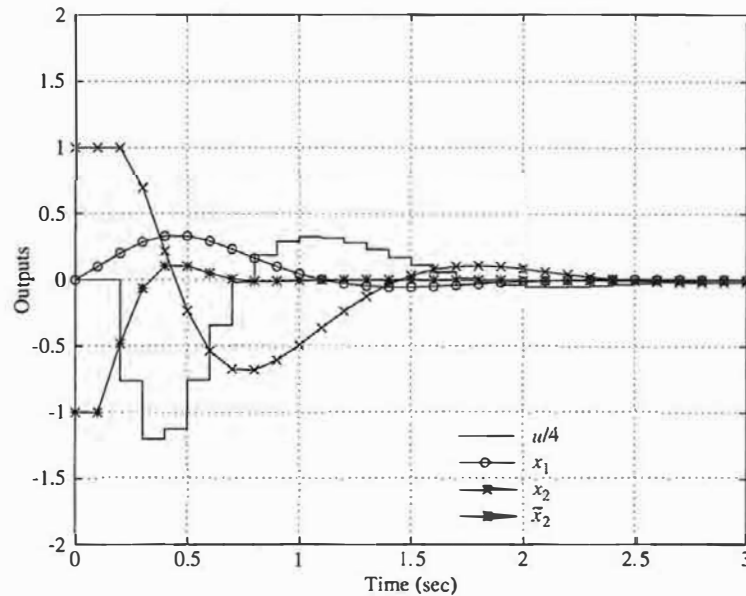
Figure 8.10 shows the response of the system and controller to the initial conditions. This could be thought of as a condition that would result from a sudden disturbance on the system. Note the estimator error at the beginning which decays in about 0.7 sec, consistent with the estimator poles. The overall system response is slower and has a settling time of about 2.5 sec, consistent with the control poles.

◆ Example 8.9 Compensation Based on the Current Estimator

Put together the full feedback control system based on the calculations already done in Examples 8.1 and 8.5, that is, using the current estimator. Use the control gain, $\mathbf{K} = [10 \ 3.5]$, and the estimator gain, $\mathbf{L}_c^T = [0.68 \ 5.2]$. Determine the $D(z)$ for comparison with a classical lead compensation. Plot the response of the system variables for an initial plant velocity of -1 rad/sec and zero for all other initial conditions. Comment on whether the responses are consistent with your expectations.

¹¹ Equation (8.60) includes a minus sign because it is the transfer function from $Y(s)$ to $U(s)$ rather than from $E(s)$, as is the normal convention used in Chapter 7.

Figure 8.10
Time histories of
controlled system with
prediction estimator,
Example 8.8



Solution. The compensation equations consist of Eq. (8.57) and, with the values of $\mathbf{K}$ and $\mathbf{L}_c$ plugged in, we find that

$$D_c(z) = -25.1 \frac{z(z - 0.792)}{z - 0.265 \pm j0.394} \quad (8.66)$$

This compensation is very much like that from the prediction estimator; however, because of the extra z in the numerator, there is no 1 cycle delay between input and output. This faster cycle response required less lead from the compensation, as exhibited by the zero being further from $z = 1$.

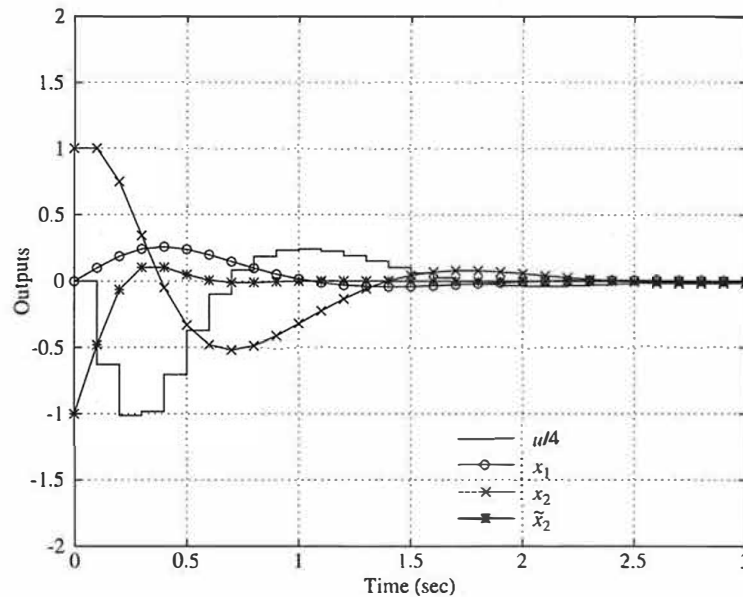
Figure 8.11 shows the response of the controlled system to the initial conditions. Note the somewhat faster response as compared to Fig. 8.10 due to the more immediate use of the measured signal.

◆ Example 8.10 Compensation Based on the Reduced-Order Estimator

Put together the full feedback control system based on the calculations already done in Examples 8.1 and 8.6, that is, using the reduced-order estimator. Use the control gain, $\mathbf{K} = [10 \ 3.5]$, and the estimator gain, $\mathbf{L}_r^T = 5$. Determine the $D(z)$ for comparison with a classical lead compensation. Plot the response of the system variables for an initial plant velocity of -1 rad/sec and zero for all other initial conditions. Comment on whether the responses are consistent with your expectations.

Also, use the $D(z)$ to construct a root locus for this system and show where the desired root locations for the control and estimation lie on the locus.

Figure 8.11
Time histories of
controlled system with
current estimator,
Example 8.9



Solution. The compensation equations consist of Eqs. (8.50) and (8.61). With the values of $\mathbf{K}$ and $\mathbf{L}$, plugged in, we find after much algebra that

$$D_r(z) = -27.7 \frac{z - 0.8182}{z - 0.2375}. \quad (8.67)$$

Figure 8.12 shows the response of the system to the initial conditions. It is very similar to that of Example 8.9; the only notable difference is the first-order response of the estimator error, which slightly reduced the control usage.

This compensation now looks exactly like the classic lead compensation that was used often in Chapter 7 and would typically be used for a $1/s^2$ plant. A sketch of a root locus vs. K is given in Fig. 8.13. For this design a gain of 27.7 is now the variable K .

The closed-loop root locations corresponding to $K = 27.7$ are indicated by the triangles and lie on the two control roots at $z = 0.8 \pm j0.25$ and on the one estimator root at $z = 0.5$, as they should.

The higher order compensators that resulted in Examples 8.8 and 8.9 have the benefit of more attenuation at high frequencies, thus reducing the sensitivity to measurement noise. This also follows from the fact that they provided a smoothed value of the measured output as well as reconstructing the velocity state variable. Full order estimators are also easier to implement because of the simpler matrix equations that result using $u = -\mathbf{K}\hat{\mathbf{x}}$ rather than the partitioned Eq. (8.61). As a result, reduced-order estimation is not often used in practice.

Figure 8.12
Time histories of system
with reduced-order
estimator, Example 8.10

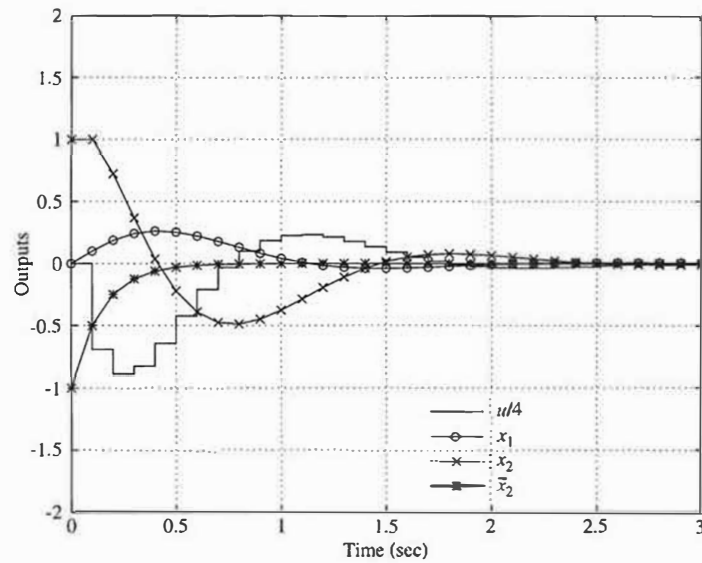
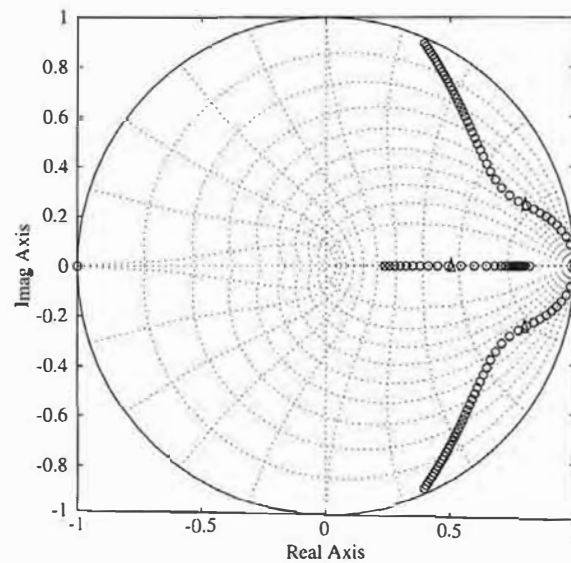


Figure 8.13
Sketch of the root locus
for Example 8.10



8.3.2 Guidelines for Pole Placement

The selection criteria of the closed-loop control and estimator poles (or roots) have been encountered throughout the examples in Chapter 8 to this point. Also see the review of state-space design for continuous systems in Section 2.6 as well

as Franklin, Powell, and Emami-Naeini (2019), Sections 7.6 and 7.7.3. The key idea for control-pole selection is that one needs to pick the poles so that the design specifications are met while the use of control is kept to a level that is no more than needed to meet the specifications. This pole-selection criterion will keep the actuator sizes to a minimum, which helps to minimize the cost and weight of the control system. The relationships between various system specifications developed in Section 7.1 can be used as an aid in the pole-selection process. For high-order systems, it is sometimes helpful to use the ITAE or Bessel prototype design root locations as discussed in Section 7.6 in Franklin, Powell, and Emami-Naeini (2019). For the case where there is a lightly damped open-loop mode, a technique that minimizes control usage is simply to add damping with little or no change in frequency, a technique called **radial projection** that was demonstrated in Example 8.3.

The optimal design methods discussed in Chapter 9 can also be used to select pole locations. They are based on minimizing a cost function that consists of the weighted sum of squares of the state errors and control. The relative weightings between the state errors and control are varied by the designer in order to meet all the system specifications with the minimum control. Optimal methods can be applied to the SISO systems, which are the subject of this chapter, or to MIMO systems.

Estimator-error pole selection is a similar kind of design process to the control-pole selection process; however, the design trade-off is somewhat different. Fast poles in an estimator do not carry the penalty of a large actuator like they do in the control case because the large signals exist only in the computer. The penalty associated with fast estimator poles is that they create an increased sensitivity between sensor errors and estimation errors.

The key idea for estimator-error pole selection is that the estimation errors should be minimized with respect to the prevailing system disturbances and sensor noise. It is also convenient to keep the estimator poles faster than the control poles in order that the total system response is dominated by the control poles. Typically, we select well-damped estimator poles that are two to six times faster than the control poles in order to provide a response dominated by the control poles. For cases where this criterion produces estimation errors due to sensor noise that are unacceptably large, the poles can be slowed down to be less than two times the control poles; however, in this case the total response could be strongly influenced by the location of the estimator poles, thus coupling the estimator design with the control design and complicating the process.

In the optimal estimation discussion in Chapter 9 we will see that the optimal estimator error poles are proportional to the ratio between the plant model errors and the sensor errors. For an accurate plant model with small disturbances but large sensor errors, the optimal estimation is achieved with very low estimator gains (slow response) because the estimator is best served by relying primarily on the plant model. On the other hand, a system with a plant model that includes the possibility of large disturbances but with an accurate sensor achieves the best

estimation by using a large estimator gain (fast response) in order to use the sensed information to correct the model errors as quickly as possible.

8.4 Introduction of the Reference Input

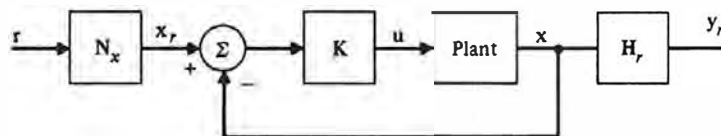
The compensation obtained by combining the control law studied in Section 8.1 with any of the estimators of Section 8.2 is a regulator design in that the goal was to drive all states to zero. We designed the characteristic equations of the control and the estimator to give satisfactory natural mode transients to initial conditions or disturbances, but no mention was made of how to structure a reference input or of the considerations necessary to obtain good transient response to reference inputs. To study these matters we will consider first how one introduces a reference input to the full-state feedback case, and then we will proceed to the case with estimators. We then turn to a discussion of output error command, a structure that occurs when the sensor is capable of providing only an error signal, for example, an attitude error from a gyro or the pointing error from a radar signal. The output error structure is also the one that results if one is designing the compensation using transfer-function methods and the reference input is structured according to Fig. 7.5, the typical case. It is, therefore, of interest to study this structure in order to understand the impact of its use on the dynamic response of the system. In conclusion, we will discuss the implications of this section's results and compare the relative advantages of the structure made possible by the state-space control/estimation approach with the classical approach.

8.4.1 Reference Inputs for Full-State Feedback

Let us first consider a reference input for a full-state feedback system as in Eq. (8.5). The structure is shown in Fig. 8.14 and consists of a state command matrix N_x that defines the desired value of the state, x_r . We wish to find N_x so that some system output, $y_r = H_r x$, is at a desired reference value. This desired output, y_r , might not be the same quantity that we sense and feed to an estimator that has been called y and determined by H in the previous two sections.

Although so far in this book we have only considered systems with a single control input and single output (SISO), Chapter 9 considers the case of more

Figure 8.14
Block diagram for
full-state feedback with
reference input



than one input and output (MIMO), and, therefore, we will allow for that in the development here and in the following subsection. We will, however, require that the number of inputs in $\mathbf{u}$ and desired outputs in $\mathbf{y}_r$ be the same.¹² The basic idea in determining $\mathbf{N}_x$ is that it should transform the reference input, $\mathbf{r}$, to a reference state that is an equilibrium one for that $\mathbf{r}$. For example, for a step command to Example 8.1, $\mathbf{N}_x^T = [1 \ 0]$; that is, we wish to command the position state element, but the velocity state element will be zero in steady state. For the double mass-spring system of Example 8.3, if we desire that $\mathbf{d} = \mathbf{r}$ —that is, that $\mathbf{H}_r = [1 \ 0 \ 0 \ 0]$ —then we should set $\mathbf{N}_x^T = [1 \ 0 \ 1 \ 0]$ because that provides a state reference, $\mathbf{x}_r$, that, if matched by the actual state, is at equilibrium for the desired output.

More specifically, we have defined $\mathbf{N}_x$ so that

$$\mathbf{N}_x \mathbf{r} = \mathbf{x}_r \quad \text{and} \quad \mathbf{u} = -\mathbf{K}(\mathbf{x} - \mathbf{x}_r). \quad (8.68)$$

If the system is Type 1 or higher and $\mathbf{r}$ is a step, there will be no steady-state error, and the final state

$$\mathbf{x}(\infty) = \mathbf{x}_{ss} = \mathbf{x}_r.$$

For Type 0 systems, there will be an error because some control is required to maintain the system at the desired $\mathbf{x}_r$.

Often the designer has sufficient knowledge of the plant to know what the equilibrium state is for the desired output, in which case the determination of $\mathbf{N}_x$ is complete. For complex plants, however, this can be difficult. In these cases, it is useful to solve for the equilibrium condition that satisfies $\mathbf{y}_r = \mathbf{r}$.¹³

In order for the solution to be valid for all system types, whether they require a steady-state control input or not, we will include the possibility of a steady-state control term that is proportional to the reference input step, that is,

$$\mathbf{u}_{ss} = \mathbf{N}_u \mathbf{r}, \quad (8.69)$$

as shown in Fig. 8.15(a). The proportionality constant, $\mathbf{N}_u$, will be solved for in the formulation.

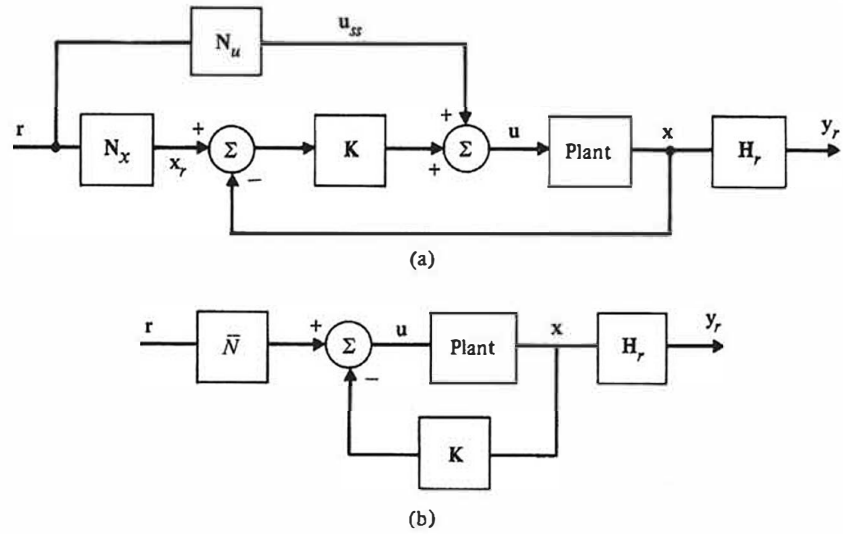
If the resulting $\mathbf{u}_{ss}$ is actually computed and implemented in the reference input structure, we refer to it as “feedforward,” but the feedforward component of the input is often not used. Instead, the preferred method of providing for zero steady-state error is through integral control or bias estimation, which essentially replaces the $\mathbf{u}_{ss}$ in Fig. 8.15(a) with an integral that is an estimate of the steady-state control, a topic that is the subject of Section 8.5. In some cases, it is difficult to achieve a high enough bandwidth when replacing feedforward with integral control; therefore, feedforward is sometimes used to reduce the demands on the integral so that it need only provide the error in the feedforward control, thus

¹² This is the only case that has a unique and exact answer, although other situations have been studied.

¹³ See Trankle and Bryson (1978) for a more complete discussion and extensions of the ideas to inputs other than steps, often called “model following.”

Figure 8.15

(a) Block diagram and
(b) modified block
diagram for full-state
feedback with reference
input and feedforward



speeding up the system response. On the other hand, if the system is Type 1 or higher, the steady-state value of control for a step will be zero, and the solution will simply give us $N_u = 0$ and N_x , which defines the desired value of the state, x_r .

Continuing on then, the steady state requirements for the system are that

$$\begin{aligned} N_x r &= x_r = x_{ss}, \\ H_r x_{ss} &= y_r = r, \end{aligned} \quad (8.70)$$

which reduce to

$$H_r N_x r = r \quad \text{and} \quad H_r N_x = I. \quad (8.71)$$

Furthermore, we are assuming the system is at steady state; therefore,

$$x(k+1) = \Phi x(k) + \Gamma u(k) \Rightarrow x_{ss} = \Phi x_{ss} + \Gamma u_{ss}$$

or

$$(\Phi - I)x_{ss} + \Gamma u_{ss} = 0;$$

and, from Eqs. (8.69) and (8.70)

$$(\Phi - I)N_x r + \Gamma N_u r = 0,$$

which reduces to

$$(\Phi - I)N_x + \Gamma N_u = 0. \quad (8.72)$$

Collecting Eqs. (8.71) and (8.72) into one matrix equation

$$\begin{bmatrix} \Phi - \mathbf{I} & \Gamma \\ \mathbf{H}_r & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{N}_x \\ \mathbf{N}_u \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix}$$

yields the desired result¹⁴

$$\begin{bmatrix} \mathbf{N}_x \\ \mathbf{N}_u \end{bmatrix} = \begin{bmatrix} \Phi - \mathbf{I} & \Gamma \\ \mathbf{H}_r & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix}. \quad (8.73)$$

It is also possible to enter the reference input after the gain multiplication according to Fig. 8.15(b) by combining $\mathbf{N}_x$ and $\mathbf{N}_u$ according to

$$\bar{\mathbf{N}} = \mathbf{N}_u + \mathbf{K}\mathbf{N}_x. \quad (8.74)$$

Calculation of $\mathbf{N}_x$ and $\mathbf{N}_u$ can be carried out by the MATLAB function `refi.m` contained in the Control Toolbox.

◆ Example 8.11 Reference Input for the Mass-Spring System

Compute the reference input quantities for the system of Example 8.3 where it is desired to command d to a new value. Compare the two structures (a) and (b) in Fig. 8.15.

Solution. The state was defined to be $\mathbf{x} = [d \quad \dot{d} \quad y \quad \dot{y}]^T$. Therefore, to command a desired value of d , $\mathbf{H}_r = [1 \quad 0 \quad 0 \quad 0]$, and the evaluation of Eq. (8.73) leads to $\mathbf{N}_x^T = [1 \quad 0 \quad 1 \quad 0]$ and $\mathbf{N}_u = 0$, as expected by inspecting the physical system shown in Fig. A.8. The fact that $\mathbf{N}_u = 0$ makes sense because this system is in equilibrium without the need of any control as long as $d = y$, which is ensured by the resulting value of $\mathbf{N}_x$. In fact, $\mathbf{N}_u$ will be zero for any system of Type 1 or higher, as already discussed. The elements of $\mathbf{N}_x$ do not depend on specific values of any parameters in the plant and are therefore not sensitive to modeling errors of the plant.

Use of Eq. (8.74) leads to $\bar{\mathbf{N}} = 0.005$ using the $\mathbf{K}$ from Eq. (8.18) and $\bar{\mathbf{N}} = 0.011$ using the $\mathbf{K}$ from Eq. (8.19). However, note that this input structure can be very sensitive to errors in $\mathbf{K}$. In this particular example, $\bar{\mathbf{N}}$ is the result of a difference of two numbers (K_1 and K_3) that are close to each other in absolute value, extremely close for the first case (which also exhibited poor response), thus producing an extreme sensitivity. Specifically, if one of the elements of $\mathbf{K}$ in Eq. (8.18) were in error by 1%, the resulting error in $\bar{\mathbf{N}}$ would be 120%! To avoid the high sensitivity for cases like this, it is advisable to structure the reference input as in Fig. 8.15(a).

This example shows that there are some systems where it is better to use the structure of Fig. 8.15(a). However, most cases do not exhibit this sensitivity and Fig. 8.15(b) is preferred due to its simplicity.

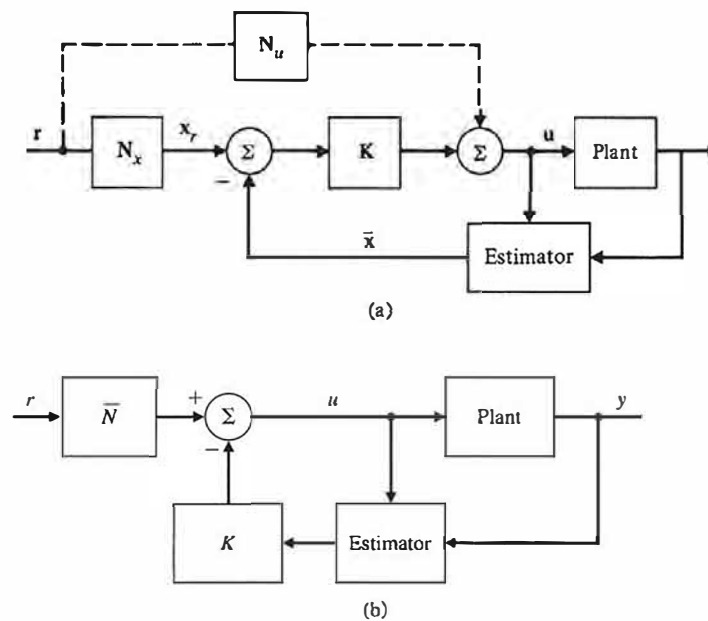
¹⁴ This cannot be solved if the plant has a zero at $z = 1$.

If the system in the example had been Type 0, we would have found that N_u was nonzero and that its value was inversely proportional to the plant gain. However, the plant gain can vary considerably in practice; therefore, designers usually choose not to use any feedforward for Type 0 systems, relying instead on the feedback to keep errors acceptably small or by implementing integral control as discussed in Section 8.5. If this is the desired action for a control design using state space, the designer can simply ignore the computed value of N_u and solely rely on N_x for guidance on how to command the state vector to the desired values.

8.4.2 Reference Inputs with Estimators: The State-Command Structure

The same ideas can be applied to the case where the estimator is used to supply a state estimate for control in place of the actual state feedback. However, it is important to structure the system so that the control u that is applied to the plant is also applied to the estimator as shown in Fig. 8.16. This means that Eqs. (8.56) and (8.57) should *not* be used in the estimator box in Fig. 8.16 because they were based on control feedback that did not include the reference input. The basic idea of the estimator that resulted in the structure of Fig. 8.4 is to drive the plant model in the estimator with the *same* inputs that are applied to the actual plant,

Figure 8.16
Block diagrams for the best reference input structure with estimators: the state command structure. (a) as derived, (b) simplified



thus minimizing estimation errors. Therefore, the form of the estimator given by Eq. (8.23) should be used with $\mathbf{u}(k)$ as shown by Fig. 8.16, that is

$$\mathbf{u}(k) = -\mathbf{K}(\bar{\mathbf{x}}(k) - \mathbf{x}_r) + \mathbf{N}_u \mathbf{r} = -\mathbf{K}\bar{\mathbf{x}}(k) + \bar{\mathbf{N}}\mathbf{r}, \quad (8.75)$$

or, with the current estimator, Eq. (8.33) is used with

$$\mathbf{u}(k) = -\mathbf{K}(\hat{\mathbf{x}}(k) - \mathbf{x}_r) + \mathbf{N}_u \mathbf{r} = -\mathbf{K}\hat{\mathbf{x}}(k) + \bar{\mathbf{N}}\mathbf{r}, \quad (8.76)$$

or, for the reduced-order estimator, Eq. (8.45) should be used with

$$\mathbf{u}(k) = -[\mathbf{K}_a \quad \mathbf{K}_b] \begin{bmatrix} x_a \\ \hat{\mathbf{x}}_b \end{bmatrix} - \mathbf{x}_r + \mathbf{N}_u \mathbf{r} = -[\mathbf{K}_a \quad \mathbf{K}_b] \begin{bmatrix} x_a \\ \hat{\mathbf{x}}_b \end{bmatrix} + \bar{\mathbf{N}}\mathbf{r}. \quad (8.77)$$

Under ideal conditions where the model in the estimator is perfect and the input $\mathbf{u}$ applied to plant and estimator is identical, no estimator error will be excited. We use the feedback to the estimator through $\mathbf{y}$ only to correct for imperfections in the estimator model, input scale factor errors, and unknown plant disturbances.

To analyze the response of a system, we must combine the estimator equations with the model of the system to be controlled. It is often useful to analyze the effect of disturbances, so the system equations (8.3) are augmented to include the disturbance, w , as

$$\begin{aligned} \mathbf{x}(k+1) &= \Phi \mathbf{x}(k) + \Gamma u(k) + \Gamma_1 w(k), \\ y(k) &= \mathbf{H}\mathbf{x}(k) + Ju(k). \end{aligned} \quad (8.78)$$

For the predictor estimator, Eq. (8.62) is augmented with the input command and disturbance as follows:

$$\begin{bmatrix} \mathbf{x} \\ \bar{\mathbf{x}} \end{bmatrix}_{k+1} = \begin{bmatrix} \Phi & -\Gamma\mathbf{K} \\ \mathbf{L}_p \mathbf{H} & \Phi - \Gamma\mathbf{K} - \mathbf{L}_p \mathbf{H} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \bar{\mathbf{x}} \end{bmatrix}_k + \begin{bmatrix} \Gamma\bar{\mathbf{N}} \\ \Gamma\bar{\mathbf{N}} \end{bmatrix} r(k) + \begin{bmatrix} \Gamma_1 \\ 0 \end{bmatrix} w(k). \quad (8.79)$$

Note that the term on the right with r introduces the command input in an identical way to both the plant and estimator equations. The term on the right with w introduces the disturbance into the plant only; the estimator is unaware of it.

It may be useful to inspect the performance of the system in terms of the desired output y_r , the control u , and the estimator error $\bar{\mathbf{x}}$. This can be accomplished with the output equation

$$\begin{bmatrix} y_r(k) \\ u(k) \\ \bar{\mathbf{x}}(k) \end{bmatrix} = \begin{bmatrix} \mathbf{H}_r & 0 \\ 0 & -\mathbf{K} \\ \mathbf{I} & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \bar{\mathbf{x}}(k) \end{bmatrix}. \quad (8.80)$$

For the current estimator, the system equations are found by combining Eqs. (8.57), (8.76), and (8.78), which yields

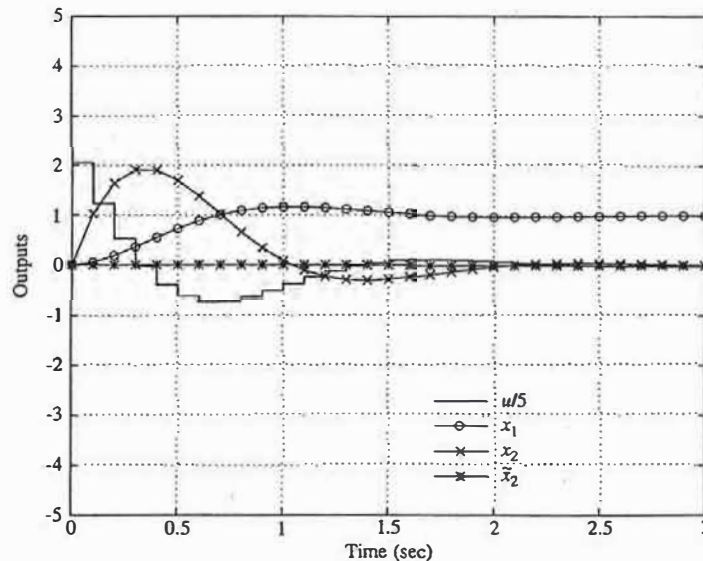
$$\begin{bmatrix} \mathbf{x} \\ \hat{\mathbf{x}} \end{bmatrix}_{k+1} = \begin{bmatrix} \Phi & -\Gamma\mathbf{K} \\ \mathbf{L}_c\mathbf{H}\Phi & \Phi - \Gamma\mathbf{K} - \mathbf{L}_c\mathbf{H}\Phi \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \hat{\mathbf{x}} \end{bmatrix}_k + \begin{bmatrix} \Gamma\bar{\mathbf{N}} \\ \Gamma\bar{\mathbf{N}} \end{bmatrix} r(k) + \begin{bmatrix} \Gamma_1 \\ \mathbf{L}_c\mathbf{H}\Gamma_1 \end{bmatrix} w(k). \quad (8.81)$$

◆ **Example 8.12** *Reference Input Command to Satellite Attitude*

Determine the state-command structure for the system whose regulator was found in Example 8.8 and verify that its step response does not excite an estimator error.

Solution. Evaluation of Eq. (8.73) yields $\mathbf{N}_x = [1 \ 0]^T$ and $N_u = 0$. Therefore, $\bar{\mathbf{N}} = \mathbf{K}_1$. The desired response of the system is obtained from Eqs. (8.79) and (8.80). Using step in MATLAB yields the unit step responses shown in Fig. 8.17. Note that the estimator error remains zero; thus the response is exactly the same as if no estimator were present. Note also that the structure shown in Fig. 8.16(b) does not allow us to represent the system in a simple, classical manner with the $D_p(z)$ from Example 8.8 placed in the upper path as in Fig. 7.5, nor does it allow us to place $D_p(z)$ in the lower feedback path. Rather, it is best to stay with the state-space description and enter the equations in the control computer based on Fig. 8.16. The response in Fig. 8.17 would also have resulted if using no estimator, a current estimator, or a reduced-order estimator.

Figure 8.17
Step-response time histories for Example 8.12



It is worthwhile to reflect on the fact that the combined system has poles that consist of the control *and* the estimator poles, as given by Eq. (8.55). The fact that the system response to an input structured as in Fig. 8.16(b) did not excite the estimator response means that the transfer function of this system had zeros that canceled the estimator poles. The determination of the structure in which the reference command is entered into a system can be viewed as one of “zero placement” and, in fact, it has been shown that it is possible to place the zeros of the closed loop transfer function at any arbitrary location [(Emami-Naeini and Franklin (1982))].

8.4.3 Output Error Command

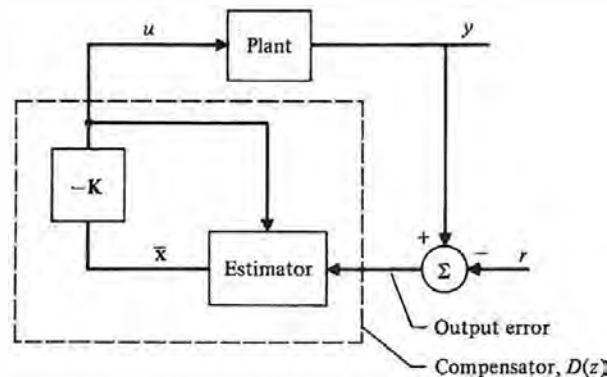
Another approach to the reference-input structure is to introduce the command only at the measured estimator input, as shown in Fig. 8.18. This solution is sometimes forced on the control designer because the sensor measures only the error. For example, many thermostats have an output that is the difference between the temperature to be controlled and the reference, or set-point, temperature. No absolute indication of the reference temperature is available to the controller. Likewise, some radar tracking systems have a reading that is proportional to the pointing error, and this signal alone must be used for control. In this case, the estimator equation (8.56) becomes

$$\bar{\mathbf{x}}(k+1) = (\Phi - \Gamma\mathbf{K} - \mathbf{L}_p\mathbf{H})\bar{\mathbf{x}} + \mathbf{L}_p(y - r),$$

and the system response can be determined by solving

$$\begin{bmatrix} \mathbf{x} \\ \bar{\mathbf{x}} \end{bmatrix}_{k+1} = \begin{bmatrix} \Phi & -\Gamma\mathbf{K} \\ \mathbf{L}_p\mathbf{H} & \Phi - \Gamma\mathbf{K} - \mathbf{L}_p\mathbf{H} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \bar{\mathbf{x}} \end{bmatrix}_k + \begin{bmatrix} \mathbf{0} \\ -\mathbf{L}_p \end{bmatrix} r(k) + \begin{bmatrix} \Gamma_1 \\ \mathbf{0} \end{bmatrix} w(k). \quad (8.82)$$

Figure 8.18
Reference input as an
output-error command



Note that the command input r only enters the estimator; therefore, the plant and estimator do not see the same command and an estimator error will be excited.

◆ **Example 8.13** *Output Command Structure with a Predictor Estimator*

Analyze the performance of Example 8.12 when using the output-command structure, specifically looking at the step response of the estimator error.

Solution. The system is analyzed using Eq. (8.82) and the output Eq. (8.80). The result is shown in Fig. 8.19. Note the estimator error response and that it causes a substantial increase in overshoot as compared to that of Fig. 8.17 (about 70% rather than 20%) and an increased use of control. Although this degradation could be reduced with faster estimator-error roots, there are limits due to the adverse effect on the estimator's sensitivity to measurement noise. Some of the degradation can be reduced by using a current or reduced order estimator because of their quicker response from the immediate use of the measured signal.

◆ **Example 8.14** *Output Command Structure with a Reduced-Order Estimator*

Analyze the performance of Example 8.13 when using the output-command structure and a reduced-order estimator.

Figure 8.19
Output time histories for
prediction estimator
with output command,
Example 8.13

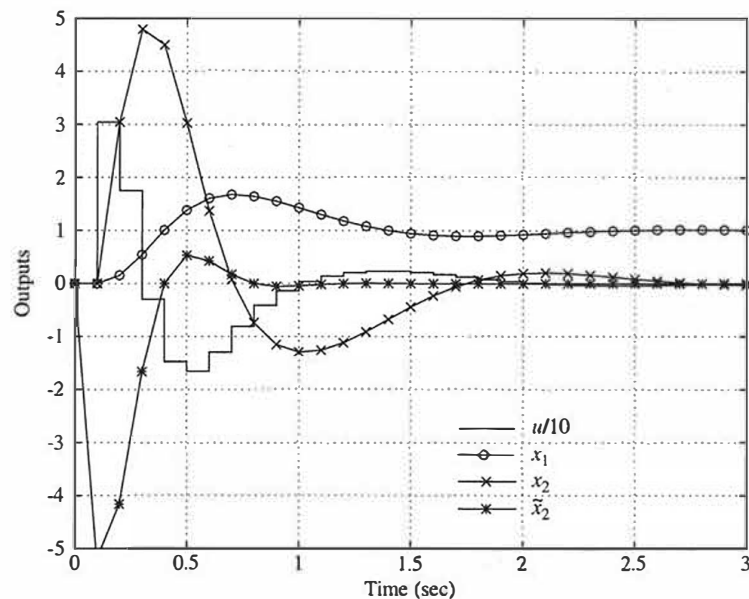
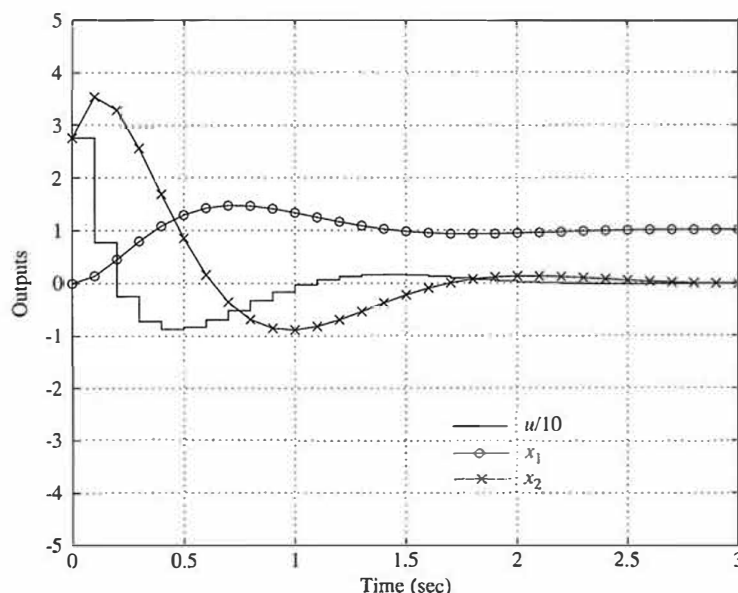


Figure 8.20

Output time histories for reduced-order estimator with output command, Example 8.14



Solution. The easiest way to analyze this system is by transform methods. The plant transfer function was analyzed in Section 4.3.1 and found to be

$$G(z) = \frac{T^2}{2} \frac{(z+1)}{(z-1)^2},$$

and the transfer function that arose from the reduced-order estimator is given by Eq. (8.67). The output time histories using these transfer functions are shown in Fig. 8.20. Note again that there is more overshoot (about 50%) than shown in Fig. 8.17 but not as much as the prediction estimator. The control usage for this case is also less than in Fig. 8.19, and it starts at the first cycle rather than being delayed by one cycle, as was the case in Fig. 8.19. This result produced a settling time for the reduced-order case that is about one sample period faster than the prediction estimator. A current estimator would produce results similar to this case because it, too, shares the feature of an immediate response to the measured output.

8.4.4 A Comparison of the Estimator Structure and Classical Methods

This section has demonstrated a key result: The controller structure based on state-space design of the control and estimator exposes a methodology for the introduction of the reference input in a way that produces a better response than that typically used with transfer-function design. The state-command input shown in Fig. 8.16 provides a faster response with less control usage than the

output-error-command scheme shown in Fig. 8.18, which is the typical transfer function structure. The reason for the advantage is that the state-command structure provides an immediate input to the plant and does not excite any modes of the compensator that degrade the response. Although this result has been specifically shown only for a simple second-order system, it applies to more complicated systems as well. In fact, for higher-order plants, higher-order compensators are often required with more modes for potential excitation.

It is not mandatory that the state-space representation and the state-estimator design approach be used in order to determine a structure that does not excite compensator modes. However, the determination is difficult using the transfer-function approach, especially in the MIMO case.

The advantages of using the transfer-function representation are that high-order systems are easier to trouble shoot, the designs are made robust with less effort, and experimental frequency-response data can be used to close a loop quickly without a time-consuming modeling effort. These advantages might not always warrant transforming to the state-space form in order to achieve easily the better reference input structure.

Although not discussed in this section using linear systems, the state-command structure allows for superior response of systems with nonlinear control characteristics, for example, saturation or on–off actuators. Whatever nonlinearity is present in the plant can usually be modeled to some degree in the estimator as well, thus reducing errors that would otherwise be excited by the nonlinearity.

◆ Example 8.15 Compensation Design for a System with a Resonance

A computer disk drive has a control system that commands the read head to move to specific tracks on the disk. One such system is described in Chapter 14. Although the dynamics between the torquer and the head motion are primarily $G(s) = 1/s^2$, there are typically several resonances due to the arm's flexibility that limit the performance of the servo system. Here we wish to limit the flexibility to one resonance mode for simplicity. The transfer function is the same as the mass-spring system shown in Appendix A.4, although the derivation of it would involve rotational dynamics rather than the linear dynamics in Appendix A.4. (See Franklin, Powell, and Emami-Naeini (2019), Chapter 2, for more details.)

The system transfer function is

$$G(s) = \frac{1 \times 10^8}{s^2(s^2 + 2\zeta_r\omega_r s + \omega_r^2)}$$

where the resonance frequency $\omega_r = 1$ kHz and the damping $\zeta_r = 0.05$. Use a sample rate of 6 kHz and design control systems that have a rise time of 10 msec with an overshoot less than 15%.

- (a) Do the design using a state estimator and the state command structure,
- (b) evaluate the $\mathbf{K}$ and $\mathbf{L}$ from (a) using the output error structure whether or not they meet the specifications, and

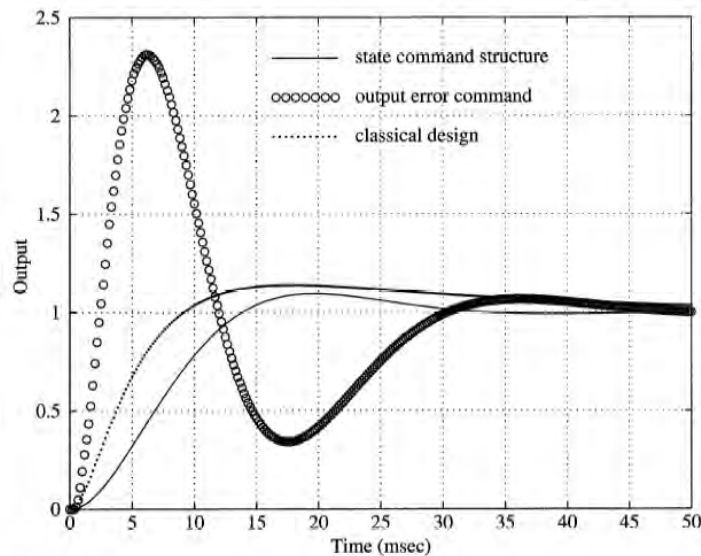
- (c) do a classical design with a lead compensator.

Iterate on whatever design parameters are appropriate for (a) and (c) to meet the design specifications.

Solution. Equation (2.16) indicates that a $t_r < 10$ msec would be met if $\omega_n > 180$ rad/sec and the $M_p < 15\%$ would be met if $\zeta > 0.5$.

- (a) For the state space design, a good starting point would be with two of the desired poles at $\omega_n = 200$ rad/sec and $\zeta = 0.6$. Two more poles also need to be selected for this 4th order system; so let's pick them at $\omega = 1$ kHz and $\zeta = 0.06$. As discussed in Section 2.6, it is not wise to move poles any more than you have to. In this case, we have retained the same natural frequency and increased the damping slightly. For the estimator, a good starting point is to pick the natural frequencies twice those of the controller, that is, at 400 rad/sec and 2 kHz. The damping was selected to be 0.7 for both sets of poles in the estimator. The selected poles were converted to their discrete equivalents and used with acker to find **K** and **L**. Evaluation of the response for the state command structure is found using Eq. (8.79) in step and is shown in Fig. 8.21. We see that the rise time and overshoot specifications were met for this case and that the system settles to its final value at about 35 msec.
- (b) Using Eq. (8.82) shows the response of the system with the same **K** and **L** for the output error command case and it is shown in the figure also. Note that the estimator excitation for this case caused an overshoot of 130% rather than the desired 15% and the system appears to be lightly damped even though its poles are in precisely the same place as case (a). It is interesting that the large overshoot was, in part, caused by using desired poles that increased the resonant mode damping from $\zeta = 0.05$ to 0.06. Had the damping

Figure 8.21
Disk read head response,
Example 8.15



been kept at 0.05, the overshoot would have been about 50% which is still excessive but significantly less. This illustrates the sensitivity of the system to the pole placement when using the output error command structure.

- (c) For the classical design, we note that the sample rate of 6 kHz is over 100 times faster than the desired closed loop natural frequency ($200 \text{ rad/sec} \cong 30 \text{ Hz}$). Therefore, the most expedient design method is to use the s -plane and convert the result to the discrete case when through. Furthermore, since the resonance frequency is significantly faster than the desired bandwidth ($\cong 200 \text{ rad/sec}$), we can ignore the resonance for the first cut at the design and simply find a compensation for a $1/s^2$ system. Using frequency response ideas, we know that a lead compensation with a ratio of 25 between the zero and pole will yield a maximum increase in phase of about 60° (Fig. 2.17). We also know that a 60° PM (Section 2.4.4) will translate to a damping ratio of about 0.6 which will meet the overshoot specification. For a $1/s^2$ plant, the phase is 180° everywhere; therefore, the desired PM will be obtained if we place the lead compensation so that the maximum phase lead is at the desired crossover point ($\cong 200 \text{ rad/sec}$). This is accomplished by placing the zero a factor of 5 below 200 (at 40 rad/sec) and the pole a factor of 5 above 200 (at 1000 rad/sec), thus producing

$$D(s) = K \frac{s + 40}{s + 1000}$$

Using this with the $G(s)$ above in a Bode plot shows that the resonant mode does not affect the design significantly, the PM is met, and the desired crossover is achieved when $K = 8000$. To convert to the digital form, we invoke `c2d` using the matched pole-zero approach and find that

$$D(z) = 7394 \frac{z + 0.9934}{z + 0.8465}$$

The closed loop step response with $D(z)$ in the forward path is found using step and shown in Fig. 8.21. It also meets the specifications, but a slow compensator mode was excited and the settling time of this system is considerably longer than the state command structure. The advantage of this approach is that the compensator is first order while the estimator approach (a) required a 4th order compensation.

8.5 Integral Control and Disturbance Estimation

Integral control is useful in eliminating the steady-state errors due to constant disturbances or reference input commands. Furthermore, most actual control systems are nonlinear and the input gain Γ and the state matrix Φ vary with time and/or the set point. The linear analysis which is the subject of this book pertains to perturbations about a set point of the nonlinear plant and the control u is a perturbation from a nominal value. The use of integral control eliminates the need to catalog nominal values or to reset the control. Rather, the integral term can be thought of as constantly calculating the value of the control required at the

set point to cause the error to go to zero. For these reasons, some form of integral control is typically included in most control systems. More generally, the external signals frequently include persistent deterministic components and the control engineer is required to design a controller which will force the steady-state error to be zero in the presence of such signals. A particular case is that of the disk drive servo required to follow a data track that is slightly off center so the reference signal is sinusoidal.

In the state-space design methods discussed so far, no mention has been made of integral control; nor have any of the design examples produced a compensation with an integral kind of behavior. In fact, state-space designs will not produce an integral action unless special steps are taken. There are two basic methods to force zero steady-state error in the presence of persistent signals: state augmentation also called internal signal model control and disturbance estimation. The idea of state augmentation for constant commands or disturbances was discussed for continuous systems in Section 2.6.5 and is essentially the same as the addition of an integral term that was discussed for transform design in Section 2.2.3. The generalization augments the state in a way that achieves zero steady-state error for a general class of reference and disturbance signals. Disturbance estimation provides the same effect based on estimation of the state of a model which could generate the external signals. We begin with the heuristic introduction of integral control.

8.5.1 Integral Control by State Augmentation

The idea is to add an integrator so as to obtain an integral of the error signal. This integrator will be physically implemented as part of the controller equations. We then feed back that integral along with the estimated or measured state as shown in Figs. 8.22 and 8.23 in a similar manner as before. To accomplish the design of the feedback gains for both the integral and the original state vector, we augment

Figure 8.22
Block diagram for
integral control with
full-state feedback

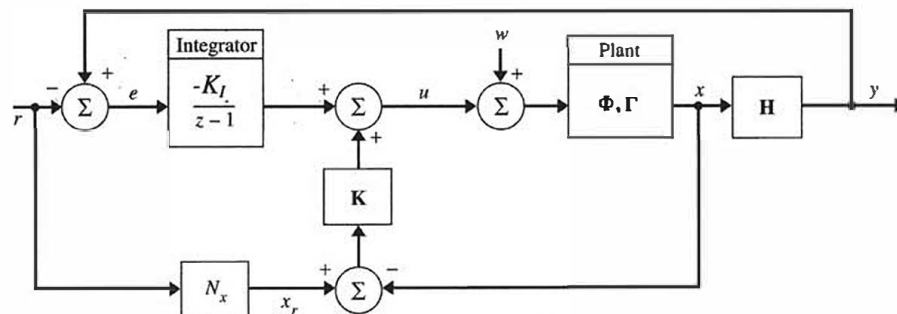
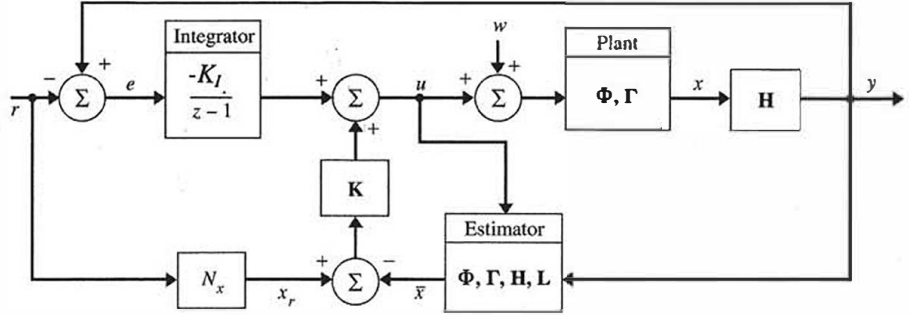


Figure 8.23
Block diagram for
integral control with
state estimation



the model of the plant with an integrator, thus adding an error integral output to the existing plant state output. This augmented model is then used as before to calculate the feedback control gains for the augmented state. More specifically, to the standard system

$$\begin{aligned}\mathbf{x}(k+1) &= \Phi\mathbf{x}(k) + \Gamma u(k) + \Gamma_1 w(k), \\ y(k) &= \mathbf{H}\mathbf{x}(k),\end{aligned}$$

we augment the state with x_I , the integral of the error, $e = y - r$. The discrete integral is simply a summation of all past values of $e(k)$ (Eq. 3.15), which results in the difference equation

$$x_I(k+1) = x_I(k) + e(k) = x_I(k) + \mathbf{H}\mathbf{x}(k) - r(k), \quad (8.83)$$

therefore arriving at the augmented plant model

$$\begin{bmatrix} x_I(k+1) \\ \mathbf{x}(k+1) \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{H} \\ 0 & \Phi \end{bmatrix} \begin{bmatrix} x_I(k) \\ \mathbf{x}(k) \end{bmatrix} + \begin{bmatrix} 0 \\ \Gamma \end{bmatrix} u(k) - \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix} r(k). \quad (8.84)$$

The control law, following Eq. (8.75), is

$$u(k) = -[K_I \quad \mathbf{K}] \begin{bmatrix} x_I(k) \\ \mathbf{x}(k) \end{bmatrix} + \mathbf{K}\mathbf{N}_x r(k).$$

With this revised definition of the system, the design techniques already developed can be used directly for the control law design. Following Figs. 8.15(a) and 8.16(a), it would be implemented as shown in Fig. 8.22 for the full-state feedback case and as shown in Fig. 8.23 for the case where an estimator is used to provide $\hat{\mathbf{x}}$. The integral is replacing the feedforward term, $\mathbf{N}_u$, and has the additional role of eliminating errors due to w .

The estimator is based on the unaugmented model and is used to reconstruct the unaugmented state. It will be n th-order, where n is the order of the original system and requires the placement of n poles. On the other hand, the design of

$[K, \mathbf{K}]$ requires the augmented system matrices (Eq. 8.84); therefore, there will be $n + 1$ poles to be selected for this portion of the design.

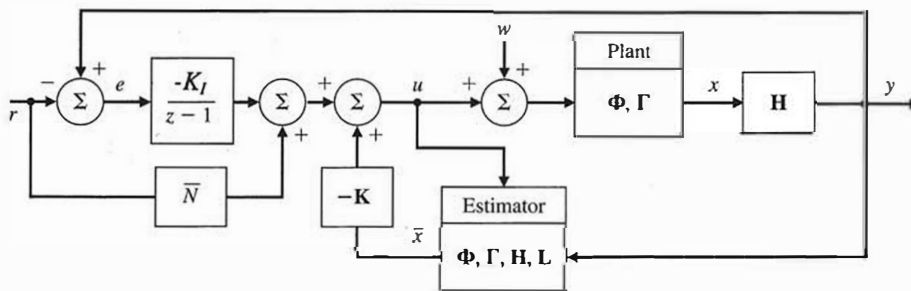
If implemented as in Fig. 8.23, the addition of the extra pole for the integrator state element will typically lead to a deteriorated command input response compared to that obtained without integral control. While it is possible to iterate on the $n + 1$ selected control poles until a satisfactory response is obtained, it is also possible to retain the reference response obtained from a non-integral-control design by placing a zero in the controller as shown in Fig. 8.24 so that it cancels the extra pole from the integrator. Note that the feedforward $\mathbf{N}_x$ term in Fig. 8.22 has been replaced by $\bar{\mathbf{N}}$ which introduces a zero at $z_i = 1 - \frac{K_f}{\bar{\mathbf{N}}}$ ¹⁵. Using the zero to cancel the closed-loop pole that was added for the integral state element cancels the excitation of that pole by command inputs. Note that this does not cancel the integral action, it merely eliminates the excitation of the extra root by command inputs. A change in the disturbance, w , will also excite the integral dynamics and the steady-state errors due to either constant disturbances or constant command inputs are eliminated. As always, the integrator output changes until its input, which is constructed to be the system error, is zero. The configuration of Fig. 8.24 can be changed to replace the feedforward of r to additional feedforward of e and modified feedback of y .

integrator pole
cancellation

◆ Example 8.16 Integral Control for the Satellite Attitude Case

Determine the integral control structure and gains for the satellite attitude control problem using full state feedback. Place the control poles at $z = 0.8 \pm j0.25, 0.9$ and use a sample

Figure 8.24
Block diagram for integral control with full-state feedback and an added zero



¹⁵ In fact, the system of Fig. 8.22 has a zero at $1 - \frac{K_f}{\bar{\mathbf{N}}}$, so the selection of the zero location corresponds to a particular selection of $\bar{\mathbf{N}}$.

period of $T = 0.1$ sec. Compute the time responses for a unit step in r at $t = 0$ sec and a step disturbance of 5 deg/sec^2 at $t = 2$ sec for (a) no integral control, (b) integral control as in Fig. 8.22, and (c) integral control as in Fig. 8.24 with the added zero at $z = +0.9$.

Solution.

- (a) This case is the same controller as used in Example 8.12. The only difference is that there is a step in the disturbance at 2 sec. Therefore, `lsim` must be used in order to allow the multiple inputs, the step in r at $t = 0$ and the disturbance step in w starting at 2 sec. The result is shown in Fig. 8.25. We see that the system responds identically to Fig. 8.17 for the first 2 sec, then there is a steady-state error in the output, x_1 , after the transients from the disturbance die out. The output error can be shown via the final value theorem to be $w/K_1 = 0.5$, thus the final value of x_1 is 1.5 instead of the commanded value of 1.
- (b) A steady-state error resulting from a disturbance is a classic motivation for the addition of integral control. The system model from Example 8.1 is augmented according to Eq. (8.84) and used with `acker` to obtain the augmented feedback gain matrix, $[K, \mathbf{K}] = [1.025 \quad 13.74 \quad 4.313]$ by asking for control roots at $z = 0.8 \pm j0.25, 0.9$. We saw from Example 8.12 that $\mathbf{N}_x = [1 \quad 0]^T$; therefore, the system design is complete and can be implemented according to Fig. 8.22. Use of `lsim` produces the response in Fig. 8.26(a). Note that the desired result has been obtained in that there is no longer a steady-state error in the output, x_1 . However, also note that it has come with a price; the behavior before the disturbance step has been degraded. More control was used, and the initial overshoot has increased from the original 20% to about 40% because of the additional root at $z = 0.9$.
- (c) The implementation is structured as shown in Fig. 8.24 which produces a zero at $z = 0.9$. All other parameters are the same as (b). Note that the resulting response in Fig. 8.26(b)

Figure 8.25
Response of satellite example to a unit reference input at $t = 0$ and a step disturbance at $t = 2$ sec with no integral control action, Example 8.16

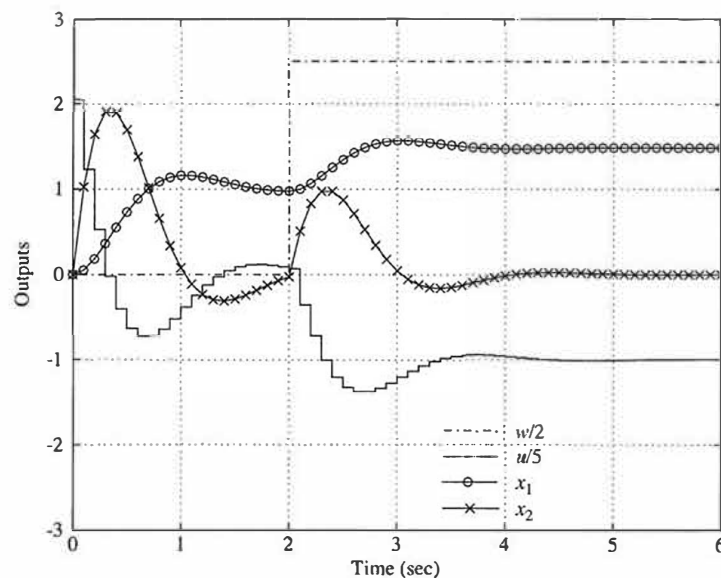
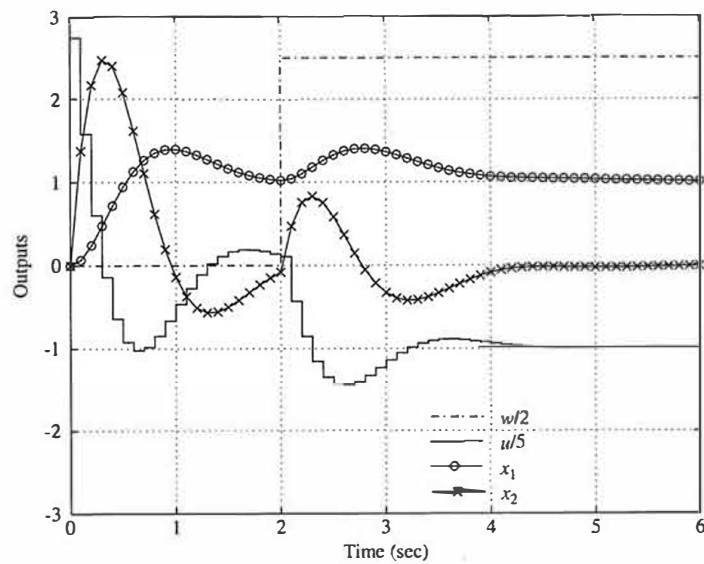
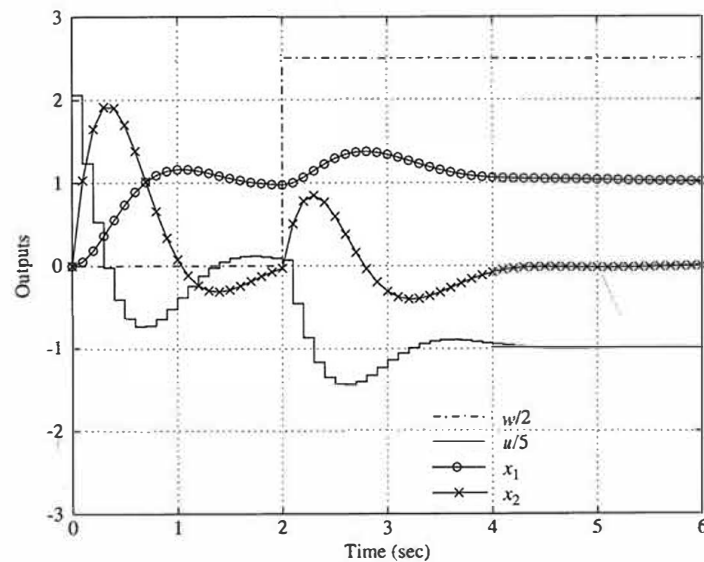


Figure 8.26

Response of satellite attitude to a unit step input at $t = 0$ and a step disturbance at $t = 2$ sec with integral control, Example 8.16. (a) As in Fig. 8.22, (b) with an added zero as in Fig. 8.24



(a)



(b)

before 2 sec is now identical to the case in Example 8.12 with no integral control; yet, the integral action successfully eliminates the steady-state error.

It should be clear from the discussion and the example that the preferred implementation of integral control is given by Fig. 8.24 where the zero cancels the integrator closed-loop root.

8.5.2 Disturbance Estimation

An alternate approach to state augmentation is to estimate the disturbance signal in the estimator and then to use that estimate in the control law so as to force the error to zero as shown in Fig. 8.27. This is called **disturbance rejection**.

disturbance rejection

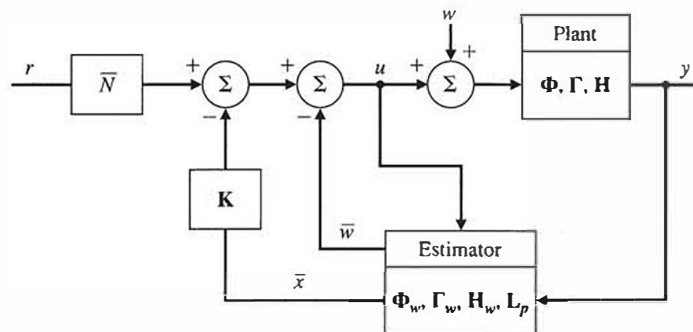
This approach yields results that are equivalent to integral control when the disturbance is a constant. After the estimate, $\bar{w}$, converges, the feedback of its value as shown in Fig. 8.27 will cancel the actual disturbance, w , and the system will behave in the steady state as if no disturbance were present. Therefore, the system will have no steady-state error, assuming, of course, that the steady-state error was due to a disturbance described by the assumed equation used in the estimator. It is important to notice that while a disturbance may, in general, appear at any point in the plant equations, the control can apply a signal to cancel it only at the control input. To reconcile these facts, we introduce the "input equivalent" disturbance. This is a virtual signal applied at the control input which would produce the *same steady state output* at y as the actual disturbance does. Then, when the control applies the negative of the virtual disturbance, the effect of the real disturbance at the output is cancelled and the error is driven to zero. To obtain an estimate of the virtual disturbance, we build the estimator with the equations of the virtual disturbance included.

disturbance modeling

Disturbances other than constant biases can be modeled, included in the estimator equations, estimated along with the plant state, and their effect on errors eliminated in steady-state. If we assume the disturbance is a constant, the continuous model is quite simple:

$$\dot{w} = 0.$$

Figure 8.27
Block diagram for input
disturbance rejection



A sinusoidal disturbance would have the model

$$\ddot{w} = -\omega_o^2 w,$$

or, in general, we could say that the disturbance obeys

$$\begin{aligned}\dot{\mathbf{x}}_d &= \mathbf{F}_d \mathbf{x}_d \\ w &= \mathbf{H}_d \mathbf{x}_d.\end{aligned}$$

and the discrete model is given by

$$\mathbf{x}_d(k+1) = \Phi_d \mathbf{x}_d(k) \quad (8.85)$$

$$w(k) = \mathbf{H}_d \mathbf{x}_d(k) \quad (8.86)$$

where $\Phi_d = e^{\mathbf{F}_d T}$. For purposes of disturbance estimation, we augment the system model with the disturbance model, so Eqs. (8.85) and (8.86) become

$$\begin{bmatrix} \mathbf{x}(k+1) \\ \mathbf{x}_d(k+1) \end{bmatrix} = \begin{bmatrix} \Phi & \Gamma_1 \mathbf{H}_d \\ \mathbf{0} & \Phi_d \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_d(k) \end{bmatrix} + \begin{bmatrix} \Gamma \\ \mathbf{0} \end{bmatrix} u(k). \quad (8.87)$$

$$y = \begin{bmatrix} \mathbf{H} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_d \end{bmatrix}, \quad (8.88)$$

which can be written as

$$\begin{aligned}\begin{bmatrix} \mathbf{x}(k+1) \\ \mathbf{x}_d(k+1) \end{bmatrix} &= \Phi_w \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_d(k) \end{bmatrix} + \Gamma_w u(k) \\ y(k) &= \mathbf{H}_w \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_d(k) \end{bmatrix}\end{aligned}$$

In the particular case where the disturbance is a constant, these equations reduce to

$$\begin{bmatrix} x(k+1) \\ w(k+1) \end{bmatrix} = \begin{bmatrix} \Phi & \Gamma_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x(k) \\ w(k) \end{bmatrix} + \begin{bmatrix} \Gamma \\ 0 \end{bmatrix} u(k). \quad (8.89)$$

$$y(k) = \begin{bmatrix} \mathbf{H} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ w(k) \end{bmatrix}. \quad (8.90)$$

All the ideas of state estimation in Section 8.2 still apply, and any of the estimation methods can be used to reconstruct the state consisting of $\mathbf{x}$ and $\mathbf{x}_d$, provided the system is observable.¹⁶ The computation of the required estimator gains is exactly as given in Section 8.2, the only change being that the system model is the augmented one given above by Φ_w and $\mathbf{H}_w$. Note from Fig. 8.27, however, that the control gain matrix, $\mathbf{K}$, is *not* obtained using the augmented model. Rather, it is obtained using the Φ and Γ associated with the unaugmented $\mathbf{F}$ and $\mathbf{G}$. In fact, the augmented system described by $[\Phi_w, \mathbf{H}_w]$ will always be uncontrollable! We have no influence over the value of w by means of the control

¹⁶ Observability requires that the virtual disturbance is "seen" at the plant output. Thus, if the disturbance is a constant, then the plant cannot have a zero from u to y at $z = 1$.

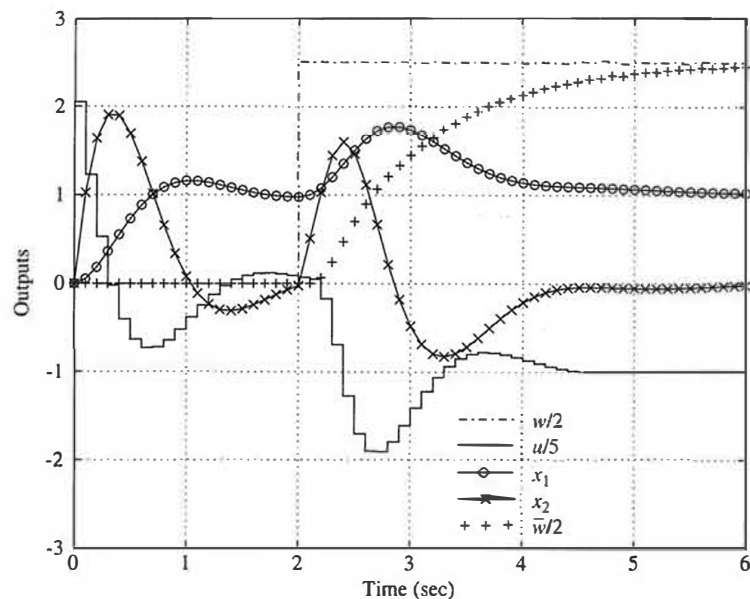
input, u , and must live with whatever value nature deals us; hence, the augmented system is uncontrollable. Our plan is not to control w , but to use the estimated value of w in a feedforward control scheme to eliminate its effect on steady-state errors. This basic idea works if w is a constant, a sinusoid, or any combination of functions that can be generated by a linear model. It works regardless of where the actual disturbance acts since the design is based on the virtual disturbance. The only constraint is that the disturbance state, $\mathbf{x}_d$, be observable.

◆ **Example 8.17** *Bias Estimation and Rejection for the Satellite Attitude Case*

Determine the bias rejection control structure and gains for the satellite attitude control problem. Place the control poles at $z = 0.8 \pm j0.25$ and use a sample period of $T = 0.1$ sec. Use a predictor estimator and place the estimator poles at $z = 0.4 \pm j0.4, 0.9$. Compute the time responses for a unit step in r at $t = 0$ sec and a step disturbance of 5 Deg/sec^2 at $t = 2$ sec and compare the results with the integral control in Example 8.16.

Solution. For purposes of finding the control gain, we use the unaugmented model as in Example 8.1 and, therefore, find the same value of $\mathbf{K} = [10.25 \ 3.49]$. For purposes of designing the estimator, we augment the plant model according to Eqs. (8.89) and (8.90) and find that the desired poles yield $\mathbf{L}_p^T = [1.3 \ 6.14 \ 5.2]$. Structuring the control as in Fig. 8.27 and applying the inputs as specified above, lsim yields the response as shown in Fig. 8.28. Note the similarity to Example 8.16 shown in Figs. 8.26(a) and (b). The disturbance rejection approach also eliminates the steady-state error. But also note the early response to the reference

Figure 8.28
Response of satellite example to a unit reference input at $t = 0$ and a step disturbance at $t = 2$ sec with bias estimation as in Fig. 8.27, Example 8.17



input: There is no increased overshoot as in Fig. 8.26(a); in fact, the response is identical to Fig. 8.26(b) up until the disturbance enters at $t = 2$ sec. Note further that the disturbance estimate, $\bar{w}$, approaches the actual disturbance value asymptotically. Notice that in this case the steady state error due to the reference input is made to be zero by the calculation of $\bar{N}$ and is not robust to small parameter changes in the way provided by integral control.

Example 8.17 shows that disturbance estimation can be used to estimate a constant disturbance input and then to use that estimate so as to reject the effect of the disturbance on steady-state errors. When the disturbance is a constant, this approach essentially duplicates the function of integral control. The following example shows how disturbance estimation can be used to estimate the value of the disturbance when it is a sinusoid. The estimate is used to cancel the effect of the disturbance, thus creating disturbance rejection.

◆ Example 8.18 *Disturbance Torque Rejection for a Spinning Satellite*

For a spinning satellite, a disturbance torque from solar pressure acts on the system as a sinusoid at the spin frequency. The attitude dynamics become 4th order as the two axes are now coupled; however, for slow spin rates, the dynamics may be approximated to be $1/s^2$ as in Example 8.17.

Determine the disturbance rejection control structure and gains for the attitude control with a disturbance torque from solar pressure of 2 deg/sec^2 where the satellite is spinning at 15 rpm. Place the control poles at $z = 0.8 \pm j0.25$ and use a sample period of $T = 0.1$ sec, as before. Use a predictor estimator and place the estimator poles at $z = 0.4 \pm j0.4, 0.9 \pm j0.1$. The location of the estimator poles corresponding to the disturbance can be selected at a relatively slow frequency as above if the sinusoidal disturbance is known to be relatively stable in magnitude and phase. By picking those estimator poles at a slow frequency, the disturbance estimate will not respond much to other higher frequency disturbances.

Plot the time history of the disturbance, the estimate of the disturbance, and the system output to verify that, in steady-state, there is no error remaining from the disturbance. Put in a step command of 1° at $t = 5$ sec to verify that the input will not excite any unwanted estimator errors. Examine the roots of the 8th order system and explain what each of them represent.

Solution. The feedback for the unaugmented system is computed as in Example 8.17 to be

$$\mathbf{K} = [10.25 \quad 3.49].$$

The disturbance acts at the control input so there is no need for the concept of the virtual disturbance. It is modeled by choosing

$$\mathbf{F}_d = \begin{bmatrix} 0 & 1 \\ -\omega_o^2 & 0 \end{bmatrix}, \quad \Gamma_1 = \Gamma \quad \text{and} \quad \mathbf{H}_d = [1 \quad 0]$$

in Eqs. (8.87) and (8.88). Use of acker with Φ_w and H_w and the desired poles results in

$$L_p^T = [1.375 \quad 6.807 \quad 8.391 \quad -6.374].$$

The time response of the system described by Fig. 8.27 is found by use of lsim where the state of the complete system consists of the augmented state as well as the estimate of the augmented state, an 8th order system. The feedback of $\bar{w}$ can be accomplished by using an augmented feedback gain $K' = [K \quad 1 \quad 0]$. Figure 8.29 shows the results. Note in the figure that the estimate takes about 4 sec to converge to the correct value and that there is a noticeable error in the output due to the disturbance until that time. The step at 5 sec has no effect on the estimate quality and therefore the response to the step is precisely as it was originally designed. Without the disturbance rejection, there would have been a steady sinusoidal error of about 0.2° superimposed on the output.

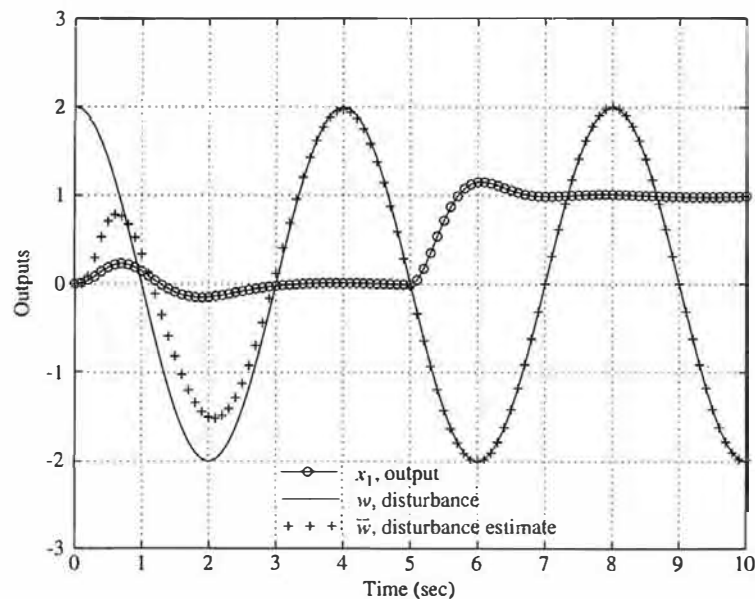
The roots of the closed loop 8th order system are:

$$z = 0.8 \pm 0.25j, \quad 0.4 \pm 0.4j, \quad 0.9 \pm 0.1j, \quad 0.988 \pm 0.156j.$$

The first 6 roots represent those selected in the control and estimation design. The last two represent the discrete equivalent of the pure oscillation at 15 rpm, which are unchanged by the feedback.

The previous example had a sinusoidal disturbance torque acting on the input to the system. It is also possible to have a measurement or sensor error that is sinusoidal in nature that one would like to eliminate as a source of error to the

Figure 8.29
Response of satellite to a sinusoidal input disturbance with disturbance rejection as in Fig. 8.27, Example 8.18



control system. The role of the estimator in this case is to use the contaminated measurement to reconstruct the error-free state for use in the control. The estimate of the sensor error is ignored by the controller.

◆ **Example 8.19** *Satellite Attitude Control with Sinusoidal Sensor Disturbance Rejection*

When an attitude sensor on a spinning satellite is misaligned from a principal axis of inertia, the sensed attitude will have a sinusoidal component as the satellite naturally spins about its principal axis. Typically, it is desirable to spin about the principal axis; therefore, it is useful to estimate the magnitude and phase of the misalignment and to reject that component of the measurement.

Repeat the design from Example 8.18, but replace the sinusoidal disturbance torque with a sinusoidal measurement error of 0.3° at 15 rpm. Again include a step command of 1° at $t = 5$ sec. Place the control poles at $z = 0.8 \pm j0.25$, as in Example 8.18, and use a sample period of $T = 0.1$ sec. Use a predictor estimator and place the estimator poles at $z = 0.8 \pm j0.2, 0.95 \pm j0.05$.

Solution. As in Example 8.18, the feedback for the unaugmented system is $\mathbf{K} = [10.25 \ 3.49]$. The output disturbance is modeled by augmenting the continuous plant with the matrices

$$\mathbf{F}_d = \begin{bmatrix} 0 & 1 \\ -\omega_o^2 & 0 \end{bmatrix}, \quad \mathbf{\Gamma}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{H}_d = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

and the augmented continuous system is

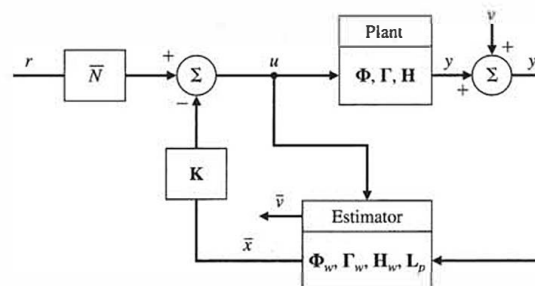
$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{F}\mathbf{x} + \mathbf{G}u \\ \dot{\mathbf{x}}_d &= \mathbf{F}_d\mathbf{x}_d \\ \mathbf{y} &= [\mathbf{H} \ \mathbf{H}_d] \end{aligned} \quad (8.91)$$

From these matrices, use of c2d will compute the discrete matrices Φ_w and $\mathbf{H}_w$ from which, with the desired poles the estimator gain is computed as

$$\mathbf{L}_p^T = [0.3899 \ 0.1624 \ 0.0855 \ 0.7378].$$

The time response of the system described by Fig. 8.30 is found by use of lsim where the state of the complete system consists of the augmented state as well as the estimate of the

Figure 8.30
Block diagram for sensor
disturbance rejection

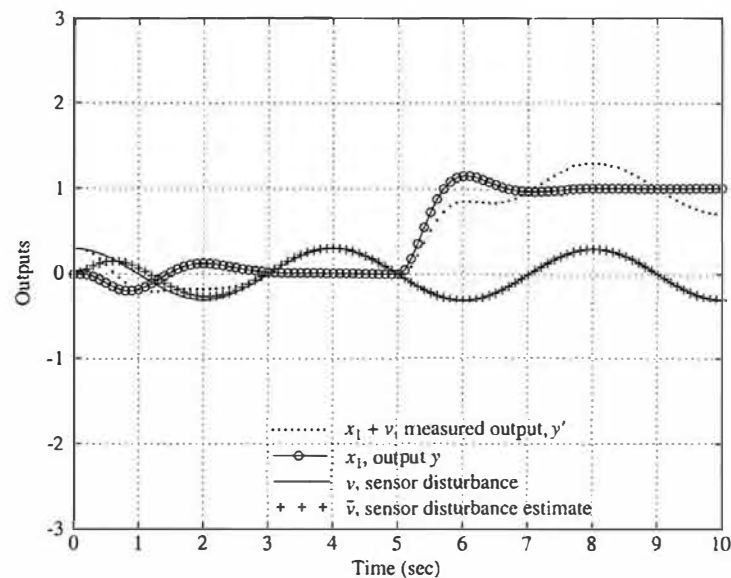


augmented state, an 8th order system. The estimate of $\bar{v}$ is ignored. Figure 8.31 shows the results. Note in the figure that the estimate takes about 3 sec to converge to the correct value and that there is a noticeable error in the output due to the measurement error until that time. The step at 5 sec has no effect on the estimate quality and therefore the response to the step is precisely as it was originally designed. Without the disturbance rejection, there would have been a steady sinusoidal error of about 0.2° as the controller attempted to follow the sinusoidal measurement.

reference following

A final example of the use of the estimator to achieve zero steady state error arises when it is desirable to track a reference signal with as little error as possible and it is known that the signal follows some persistent pattern. Since it would usually take some control effort to follow such a signal, the system would normally exhibit a following error of sufficient magnitude to produce the required control effort. This following error can be eliminated if the systematic pattern can be modeled and estimated, then used in a feedforward manner to produce the desired control effort. This is called **reference following**. The idea is the same as with disturbance rejection except that the error is now not the output only but the difference between the reference and the output. The idea again is to construct a virtual reference, ρ , at the control input which would produce the system error at the plant output, as shown in Figure 8.32(a). The feedback gain $\mathbf{K}$ is designed

Figure 8.31
Response of satellite to a sinusoidal sensor disturbance with disturbance rejection as in Fig. 8.30, Example 8.19

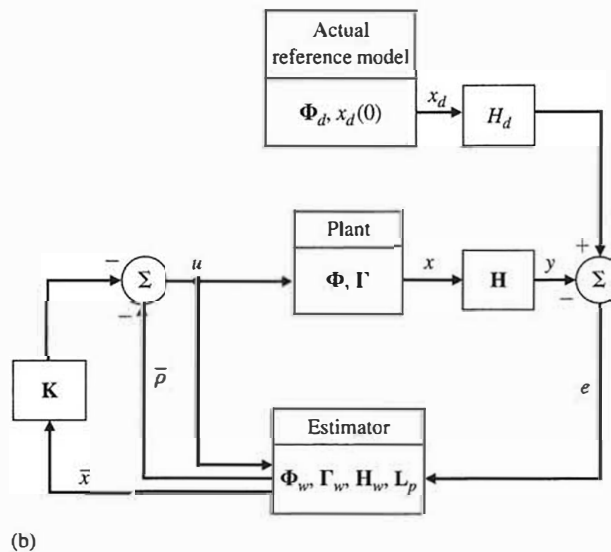
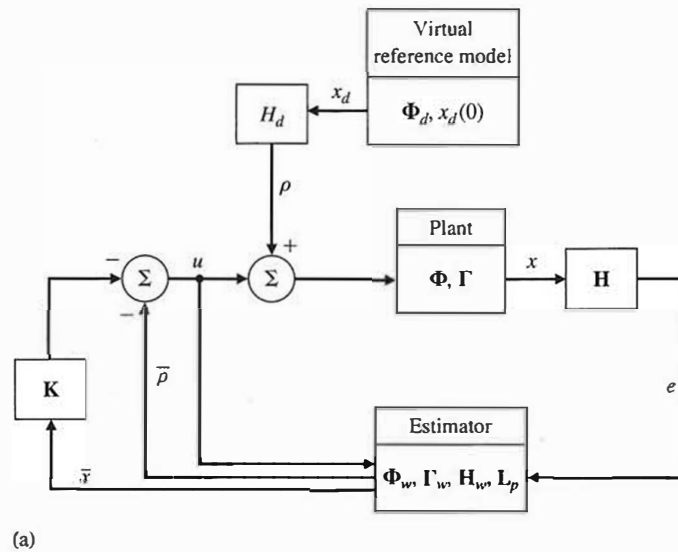


using the unaugmented plant described by Φ and Γ , but the feedback signal is the system error e .

By estimating ρ and feeding that estimate into the plant with the control as shown in Fig. 8.32(a), the effect of ρ is eliminated in this virtual model and $e \equiv 0$ in steady state. The actual situation that is implemented is shown in Fig. 8.32(b) where the reference is subtracted from the output to form the error, e , and the

Figure 8.32

Block diagram for sensor disturbance following.
(a) The pretend model,
and (b) the
implementation model



estimate of the virtual reference, $\bar{\rho}$, is subtracted from the control. Therefore, e will behave as if ρ was canceled, so that in steady state, $e \equiv 0$, implying that y is following r .

◆ Example 8.20 Sinusoidal Sensor Disturbance Following of a Disk Drive

A computer disk drive read head must follow tracks on the disk. Typically, there is a small offset between the center of the tracks and the rotational center of the disk, thus producing a wobble in the tracks that the read head must follow. Assume the transfer function between the input command and the read head response is

$$G(s) = \frac{1000}{s^2}$$

and the disk spins at 5000 rpm. Design a control system that follows tracks on the disk with no error even though they may be off center by some small amount. Pick the gains so that the system has a rise time better than 3 msec and no more than a 15% overshoot. Use a sample rate of 2 kHz.

Solution. The disk head system is described by

$$\mathbf{F} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} 0 \\ 1000 \end{bmatrix}, \quad \text{and} \quad \mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

We pretend that the disturbance is the same as in Example 8.18, that is

$$\mathbf{F}_d = \begin{bmatrix} 0 & 1 \\ -\omega_o^2 & 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{H}_d = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

where $\omega_o = 5000 \text{ rpm}$ or 523.6 rad/sec . The difference here compared to Example 8.18 is that the actual reference is being subtracted from the output and we wish to eliminate its effect on e . In other words, we want y to follow r in Fig. 8.32(b).

Using $\mathbf{F}$ and $\mathbf{G}$ with `c2d` yields Φ and Γ for the disk head dynamics. To meet the rise time and overshoot specifications, a few iterations show that poles with a natural frequency of 1000 rad/sec and a 0.6 damping ratio, converted to the digital domain, produce the desired response. The resulting feedback gain is

$$\mathbf{K} = [736.54 \quad 1.0865].$$

But we wish to include the virtual reference in the control as

$$u = \mathbf{K}\mathbf{x} - \rho.$$

The estimator is constructed using the augmented system. A satisfactory design was found with equivalent s -plane poles that had natural frequencies of 1000 and 1200 rad/sec, both sets with $\zeta = 0.9$. Use of `acker` with Φ_w and $\mathbf{H}_w$ results in $\mathbf{L}_p^T = [3.9 \quad 1012.9 \quad -2.4 \quad 553.0]$.

The time response of the system described by Fig. 8.32(b) is found by use of `lsim` where the state consists of the augmented state as well as the estimate of the augmented state, an 8th order system.

The purpose of the feedforward is to provide a control that will produce $y = r$ and thus $e_{ss} \equiv 0$. Since we have an estimate of an input, $\bar{\rho}$, that will produce the track wobble which is the tracking reference, r , feedforward of $-\bar{\rho}$ causes $y = -r$. This produces the time response